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Small form-factor devices with integrated and modular harvesting receivers, and shelving-mounted wireless-power transmitters for use therewith

Abstract

A wireless-power harvester integrated in a small device, comprising a stamped metal harvesting antenna. The stamped metal antenna is formed into a meandering shape. A first end of the meandering shape is a free end positioned within free space of a housing of a small device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small device. The PCB is configured to operate as a ground plane for the stamped metal antenna. An intermediate portion, disposed between the first end and the second end of the meandering shape, is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small device or for powering the small device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
787412	12/1904	Tesla	N/A	N/A
2811624	12/1956	Haagensen	N/A	N/A
2863148	12/1957	Gammon et al.	N/A	N/A
3167775	12/1964	Guertler	N/A	N/A
3434678	12/1968	Brown et al.	N/A	N/A
3696384	12/1971	Lester	N/A	N/A
3754269	12/1972	Clavin	N/A	N/A
4101895	12/1977	Jones, Jr.	N/A	N/A
4360741	12/1981	Fitzsimmons et al.	N/A	N/A
4944036	12/1989	Hyatt	N/A	N/A
4995010	12/1990	Knight	N/A	N/A
5142292	12/1991	Chang	N/A	N/A
5200759	12/1992	McGinnis	N/A	N/A
5211471	12/1992	Rohrs	N/A	N/A
5276455	12/1993	Fitzsimmons et al.	N/A	N/A
5548292	12/1995	Hirshfield et al.	N/A	N/A
5556749	12/1995	Mitsubishi et al.	N/A	N/A
5568088	12/1995	Dent et al.	N/A	N/A
5631572	12/1996	Sheen et al.	N/A	N/A
5646633	12/1996	Dahlberg	N/A	N/A
5697063	12/1996	Kishigami et al.	N/A	N/A
5712642	12/1997	Hulderman	N/A	N/A
5936527	12/1998	Isaacman et al.	N/A	N/A
5982139	12/1998	Parise	N/A	N/A
6046708	12/1999	MacDonald, Jr. et al.	N/A	N/A
6061025	12/1999	Jackson et al.	N/A	N/A
6127799	12/1999	Krishnan	N/A	N/A
6127942	12/1999	Welle	N/A	N/A

6163296	12/1999	Lier et al.	N/A	N/A
6176433	12/2000	Uesaka et al.	N/A	N/A
6271799	12/2000	Rief	N/A	N/A
6289237	12/2000	Mickle et al.	N/A	N/A
6329908	12/2000	Frecska	N/A	N/A
6400586	12/2001	Raddi et al.	N/A	N/A
6421235	12/2001	Ditzik	N/A	N/A
6437685	12/2001	Hanaki	N/A	N/A
6456253	12/2001	Rummeli et al.	N/A	N/A
6476795	12/2001	Derocher et al.	N/A	N/A
6501414	12/2001	Arndt et al.	N/A	N/A
6583723	12/2002	Watanabe et al.	N/A	N/A
6597897	12/2002	Tang	N/A	N/A
6615074	12/2002	Mickle et al.	N/A	N/A
6650376	12/2002	Obitsu	N/A	N/A
6664920	12/2002	Mott et al.	N/A	N/A
6680700	12/2003	Hilgers	N/A	N/A
6798716	12/2003	Charych	N/A	N/A
6803744	12/2003	Sabo	N/A	N/A
6853197	12/2004	McFarland	N/A	N/A
6856291	12/2004	Mickle et al.	N/A	N/A
6911945	12/2004	Korva	N/A	N/A
6960968	12/2004	Odendaal et al.	N/A	N/A
6967462	12/2004	Landis	N/A	N/A
6988026	12/2005	Breed et al.	N/A	N/A
7003350	12/2005	Denker et al.	N/A	N/A
7012572	12/2005	Schaffner et al.	N/A	N/A
7027311	12/2005	Vanderelli et al.	N/A	N/A
7068234	12/2005	Sievenpiper	N/A	N/A
7068991	12/2005	Parise	N/A	N/A
7079079	12/2005	Jo et al.	N/A	N/A
7183748	12/2006	Unno et al.	N/A	N/A
7191013	12/2006	Miranda et al.	N/A	N/A
7193644	12/2006	Carter	N/A	N/A
7196663	12/2006	Bolzer et al.	N/A	N/A
7205749	12/2006	Hagen et al.	N/A	N/A
7215296	12/2006	Abramov et al.	N/A	N/A
7222356	12/2006	Yonezawa et al.	N/A	N/A
7274334	12/2006	O'Riordan et al.	N/A	N/A
7274336	12/2006	Carson	N/A	N/A
7351975	12/2007	Brady et al.	N/A	N/A
7359730	12/2007	Dennis et al.	N/A	N/A
7372408	12/2007	Gaucher	N/A	N/A
7392068	12/2007	Dayan	N/A	N/A
7403803	12/2007	Mickle et al.	N/A	N/A
7443057	12/2007	Nunally	N/A	N/A
7451839	12/2007	Perlman	N/A	N/A
7463201	12/2007	Chiang et al.	N/A	N/A
7471247	12/2007	Saily	N/A	N/A
7535195	12/2008	Horovitz et al.	N/A	N/A

7614556	12/2008	Overhultz et al.	N/A	N/A
7639994	12/2008	Greene et al.	N/A	N/A
7643312	12/2009	Vanderelli et al.	N/A	N/A
7652577	12/2009	Madhow et al.	N/A	N/A
7679576	12/2009	Riedel et al.	N/A	N/A
7702771	12/2009	Ewing et al.	N/A	N/A
7786419	12/2009	Hyde et al.	N/A	N/A
7812771	12/2009	Greene et al.	N/A	N/A
7830312	12/2009	Choudhury et al.	N/A	N/A
7844306	12/2009	Shearer et al.	N/A	N/A
7868482	12/2010	Greene et al.	N/A	N/A
7898105	12/2010	Greene et al.	N/A	N/A
7904117	12/2010	Doan et al.	N/A	N/A
7911386	12/2010	Itoh et al.	N/A	N/A
7925308	12/2010	Greene et al.	N/A	N/A
7948208	12/2010	Partovi et al.	N/A	N/A
8049676	12/2010	Yoon et al.	N/A	N/A
8055003	12/2010	Mittleman et al.	N/A	N/A
8070595	12/2010	Alderucci et al.	N/A	N/A
8072380	12/2010	Crouch	N/A	N/A
8092301	12/2011	Alderucci et al.	N/A	N/A
8099140	12/2011	Arai	N/A	N/A
8115448	12/2011	John	N/A	N/A
8159090	12/2011	Greene et al.	N/A	N/A
8159364	12/2011	Zeine	N/A	N/A
8180286	12/2011	Yamasuge	N/A	N/A
8184454	12/2011	Mao	N/A	N/A
8228194	12/2011	Mickle	N/A	N/A
8234509	12/2011	Gioscia et al.	N/A	N/A
8264101	12/2011	Hyde et al.	N/A	N/A
8264291	12/2011	Morita	N/A	N/A
8276325	12/2011	Clifton et al.	N/A	N/A
8278784	12/2011	Cook et al.	N/A	N/A
8284101	12/2011	Fusco	N/A	N/A
8310201	12/2011	Wright	N/A	N/A
8338991	12/2011	Von Novak et al.	N/A	N/A
8362745	12/2012	Tinaphong	N/A	N/A
8380255	12/2012	Shearer et al.	N/A	N/A
8384600	12/2012	Huang et al.	N/A	N/A
8410953	12/2012	Zeine	N/A	N/A
8411963	12/2012	Luff	N/A	N/A
8432062	12/2012	Greene et al.	N/A	N/A
8432071	12/2012	Huang et al.	N/A	N/A
8446248	12/2012	Zeine	N/A	N/A
8447234	12/2012	Cook et al.	N/A	N/A
8451189	12/2012	Fluhler	N/A	N/A
8452235	12/2012	Kirby et al.	N/A	N/A
8457656	12/2012	Perkins et al.	N/A	N/A
8461817	12/2012	Martin et al.	N/A	N/A
8467733	12/2012	Leabman	N/A	N/A

8497601	12/2012	Hall et al.	N/A	N/A
8497658	12/2012	Von Novak et al.	N/A	N/A
8552597	12/2012	Song et al.	N/A	N/A
8558661	12/2012	Zeine	N/A	N/A
8560026	12/2012	Chanterac	N/A	N/A
8564485	12/2012	Milosavljevic et al.	N/A	N/A
8604746	12/2012	Lee	N/A	N/A
8614643	12/2012	Leabman	N/A	N/A
8621245	12/2012	Shearer et al.	N/A	N/A
8626249	12/2013	Ungari et al.	N/A	N/A
8629576	12/2013	Levine	N/A	N/A
8653966	12/2013	Rao et al.	N/A	N/A
8655272	12/2013	Saunamäki	N/A	N/A
8674551	12/2013	Low et al.	N/A	N/A
8686685	12/2013	Moshfeghi	N/A	N/A
8686905	12/2013	Shtrom	N/A	N/A
8712355	12/2013	Black et al.	N/A	N/A
8712485	12/2013	Tam	N/A	N/A
8718773	12/2013	Wills et al.	N/A	N/A
8729737	12/2013	Schatz et al.	N/A	N/A
8736228	12/2013	Freed et al.	N/A	N/A
8760113	12/2013	Keating	N/A	N/A
8770482	12/2013	Ackermann et al.	N/A	N/A
8772960	12/2013	Yoshida	N/A	N/A
8823319	12/2013	Von Novak, III et al.	N/A	N/A
8832646	12/2013	Wendling	N/A	N/A
8854176	12/2013	Zeine	N/A	N/A
8860364	12/2013	Low et al.	N/A	N/A
8897770	12/2013	Frolov et al.	N/A	N/A
8903456	12/2013	Chu et al.	N/A	N/A
8917057	12/2013	Hui	N/A	N/A
8923189	12/2013	Leabman	N/A	N/A
8928544	12/2014	Massie et al.	N/A	N/A
8937408	12/2014	Ganem et al.	N/A	N/A
8946940	12/2014	Kim et al.	N/A	N/A
8963486	12/2014	Kirby et al.	N/A	N/A
8970070	12/2014	Sada et al.	N/A	N/A
8989053	12/2014	Skaaksrud et al.	N/A	N/A
9000616	12/2014	Greene et al.	N/A	N/A
9001622	12/2014	Perry	N/A	N/A
9006934	12/2014	Kozakai et al.	N/A	N/A
9021277	12/2014	Shearer et al.	N/A	N/A
9030161	12/2014	Lu et al.	N/A	N/A
9059598	12/2014	Kang et al.	N/A	N/A
9059599	12/2014	Won et al.	N/A	N/A
9077188	12/2014	Moshfeghi	N/A	N/A
9083595	12/2014	Rakib et al.	N/A	N/A
9088216	12/2014	Garrity et al.	N/A	N/A
9124125	12/2014	Leabman et al.	N/A	N/A
9130397	12/2014	Leabman et al.	N/A	N/A

9130602	12/2014	Cook	N/A	N/A
9142998	12/2014	Yu et al.	N/A	N/A
9143000	12/2014	Leabman et al.	N/A	N/A
9143010	12/2014	Urano	N/A	N/A
9153074	12/2014	Zhou et al.	N/A	N/A
9178389	12/2014	Hwang	N/A	N/A
9225196	12/2014	Huang et al.	N/A	N/A
9240469	12/2015	Sun et al.	N/A	N/A
9242411	12/2015	Kritchman et al.	N/A	N/A
9244500	12/2015	Cain et al.	N/A	N/A
9252628	12/2015	Leabman et al.	N/A	N/A
9270344	12/2015	Rosenberg	N/A	N/A
9276329	12/2015	Jones et al.	N/A	N/A
9282582	12/2015	Dunsbergen et al.	N/A	N/A
9294840	12/2015	Anderson et al.	N/A	N/A
9297896	12/2015	Andrews	N/A	N/A
9318898	12/2015	John	N/A	N/A
9368020	12/2015	Bell et al.	N/A	N/A
9401977	12/2015	Gaw	N/A	N/A
9409490	12/2015	Kawashima	N/A	N/A
9419335	12/2015	Pintos	N/A	N/A
9419443	12/2015	Leabman	N/A	N/A
9438045	12/2015	Leabman	N/A	N/A
9438046	12/2015	Leabman	N/A	N/A
9444283	12/2015	Son et al.	N/A	N/A
9450449	12/2015	Leabman et al.	N/A	N/A
9461502	12/2015	Lee et al.	N/A	N/A
9520725	12/2015	Masaoka et al.	N/A	N/A
9520748	12/2015	Hyde et al.	N/A	N/A
9522270	12/2015	Perryman et al.	N/A	N/A
9532748	12/2016	Denison et al.	N/A	N/A
9537354	12/2016	Bell et al.	N/A	N/A
9537357	12/2016	Leabman	N/A	N/A
9537358	12/2016	Leabman	N/A	N/A
9538382	12/2016	Bell et al.	N/A	N/A
9544640	12/2016	Lau	N/A	N/A
9559553	12/2016	Bae	N/A	N/A
9564773	12/2016	Pogorelik et al.	N/A	N/A
9571974	12/2016	Choi et al.	N/A	N/A
9590317	12/2016	Zimmerman et al.	N/A	N/A
9590444	12/2016	Walley	N/A	N/A
9620996	12/2016	Zeine	N/A	N/A
9647328	12/2016	Dobric	N/A	N/A
9706137	12/2016	Scanlon et al.	N/A	N/A
9711999	12/2016	Hietala et al.	N/A	N/A
9723635	12/2016	Nambord et al.	N/A	N/A
9793758	12/2016	Leabman	N/A	N/A
9793764	12/2016	Perry	N/A	N/A
9800080	12/2016	Leabman et al.	N/A	N/A
9800172	12/2016	Leabman	N/A	N/A

9806564	12/2016	Leabman	N/A	N/A
9819230	12/2016	Petras et al.	N/A	N/A
9824815	12/2016	Leabman et al.	N/A	N/A
9825674	12/2016	Leabman	N/A	N/A
9831718	12/2016	Leabman et al.	N/A	N/A
9838083	12/2016	Bell et al.	N/A	N/A
9843201	12/2016	Leabman et al.	N/A	N/A
9843213	12/2016	Leabman et al.	N/A	N/A
9843229	12/2016	Leabman	N/A	N/A
9843763	12/2016	Leabman et al.	N/A	N/A
9847669	12/2016	Leabman	N/A	N/A
9847677	12/2016	Leabman	N/A	N/A
9847679	12/2016	Bell et al.	N/A	N/A
9853361	12/2016	Chen et al.	N/A	N/A
9853692	12/2016	Bell et al.	N/A	N/A
9859756	12/2017	Leabman et al.	N/A	N/A
9859758	12/2017	Leabman	N/A	N/A
9859797	12/2017	Leabman	N/A	N/A
9866279	12/2017	Bell et al.	N/A	N/A
9867032	12/2017	Verma et al.	N/A	N/A
9867062	12/2017	Bell et al.	N/A	N/A
9871301	12/2017	Contopanagos	N/A	N/A
9876379	12/2017	Leabman et al.	N/A	N/A
9876380	12/2017	Leabman et al.	N/A	N/A
9876394	12/2017	Leabman	N/A	N/A
9876536	12/2017	Bell et al.	N/A	N/A
9876648	12/2017	Bell	N/A	N/A
9882394	12/2017	Bell et al.	N/A	N/A
9882427	12/2017	Leabman et al.	N/A	N/A
9887584	12/2017	Bell et al.	N/A	N/A
9887739	12/2017	Leabman et al.	N/A	N/A
9891669	12/2017	Bell	N/A	N/A
9893554	12/2017	Bell et al.	N/A	N/A
9893555	12/2017	Leabman et al.	N/A	N/A
9893564	12/2017	de Rochemont	N/A	N/A
9899744	12/2017	Contopanagos et al.	N/A	N/A
9899844	12/2017	Bell et al.	N/A	N/A
9899861	12/2017	Leabman et al.	N/A	N/A
9899873	12/2017	Bell et al.	N/A	N/A
9906065	12/2017	Leabman et al.	N/A	N/A
9906275	12/2017	Leabman	N/A	N/A
9912199	12/2017	Leabman et al.	N/A	N/A
9916485	12/2017	Lilly et al.	N/A	N/A
9917477	12/2017	Bell et al.	N/A	N/A
9923386	12/2017	Leabman et al.	N/A	N/A
9939864	12/2017	Bell et al.	N/A	N/A
9941747	12/2017	Bell et al.	N/A	N/A
9941754	12/2017	Leabman et al.	N/A	N/A
9948135	12/2017	Leabman et al.	N/A	N/A
9954374	12/2017	Bell et al.	N/A	N/A

9965009	12/2017	Bell et al.	N/A	N/A
9966765	12/2017	Leabman	N/A	N/A
9966784	12/2017	Leabman	N/A	N/A
9967743	12/2017	Bell et al.	N/A	N/A
9973008	12/2017	Leabman	N/A	N/A
10003211	12/2017	Leabman et al.	N/A	N/A
10008777	12/2017	Broyde et al.	N/A	N/A
10008875	12/2017	Leabman	N/A	N/A
10008889	12/2017	Bell et al.	N/A	N/A
10014728	12/2017	Leabman	N/A	N/A
10027159	12/2017	Hosseini	N/A	N/A
10038337	12/2017	Leabman et al.	N/A	N/A
10050462	12/2017	Leabman et al.	N/A	N/A
10056782	12/2017	Leabman	N/A	N/A
10063064	12/2017	Bell et al.	N/A	N/A
10063105	12/2017	Leabman	N/A	N/A
10063106	12/2017	Bell et al.	N/A	N/A
10068703	12/2017	Contopanagos	N/A	N/A
10075008	12/2017	Bell et al.	N/A	N/A
10079515	12/2017	Hosseini et al.	N/A	N/A
10090699	12/2017	Leabman	N/A	N/A
10090714	12/2017	Bohn et al.	N/A	N/A
10090886	12/2017	Bell et al.	N/A	N/A
10103552	12/2017	Leabman et al.	N/A	N/A
10103582	12/2017	Leabman et al.	N/A	N/A
10110046	12/2017	Esquibel et al.	N/A	N/A
10116170	12/2017	Leabman	N/A	N/A
10122219	12/2017	Hosseini et al.	N/A	N/A
10122415	12/2017	Bell et al.	N/A	N/A
10124754	12/2017	Leabman	N/A	N/A
10128686	12/2017	Leabman et al.	N/A	N/A
10128693	12/2017	Bell et al.	N/A	N/A
10128695	12/2017	Leabman et al.	N/A	N/A
10128699	12/2017	Leabman	N/A	N/A
10134260	12/2017	Bell et al.	N/A	N/A
10135112	12/2017	Hosseini	N/A	N/A
10135286	12/2017	Hosseini et al.	N/A	N/A
10135294	12/2017	Leabman	N/A	N/A
10135295	12/2017	Leabman	N/A	N/A
10141768	12/2017	Leabman et al.	N/A	N/A
10141771	12/2017	Hosseini et al.	N/A	N/A
10141791	12/2017	Bell et al.	N/A	N/A
10148097	12/2017	Leabman et al.	N/A	N/A
10153645	12/2017	Bell et al.	N/A	N/A
10153653	12/2017	Bell et al.	N/A	N/A
10153660	12/2017	Leabman et al.	N/A	N/A
10158257	12/2017	Leabman	N/A	N/A
10158259	12/2017	Leabman	N/A	N/A
10164478	12/2017	Leabman	N/A	N/A
10170917	12/2018	Bell et al.	N/A	N/A

10177594	12/2018	Contopanagos	N/A	N/A
10181756	12/2018	Bae et al.	N/A	N/A
10186892	12/2018	Hosseini et al.	N/A	N/A
10186893	12/2018	Bell et al.	N/A	N/A
10186911	12/2018	Leabman	N/A	N/A
10186913	12/2018	Leabman et al.	N/A	N/A
10193396	12/2018	Bell et al.	N/A	N/A
10199835	12/2018	Leabman	N/A	N/A
10199849	12/2018	Bell	N/A	N/A
10199850	12/2018	Leabman	N/A	N/A
10205239	12/2018	Contopanagos et al.	N/A	N/A
10206185	12/2018	Leabman et al.	N/A	N/A
10211674	12/2018	Leabman et al.	N/A	N/A
10211680	12/2018	Leabman et al.	N/A	N/A
10211682	12/2018	Bell et al.	N/A	N/A
10211685	12/2018	Bell et al.	N/A	N/A
10218207	12/2018	Hosseini et al.	N/A	N/A
10218227	12/2018	Leabman et al.	N/A	N/A
10223717	12/2018	Bell	N/A	N/A
10224758	12/2018	Leabman et al.	N/A	N/A
10224982	12/2018	Leabman	N/A	N/A
10230266	12/2018	Leabman et al.	N/A	N/A
10243414	12/2018	Leabman et al.	N/A	N/A
10256657	12/2018	Hosseini et al.	N/A	N/A
10256677	12/2018	Hosseini et al.	N/A	N/A
10263432	12/2018	Leabman et al.	N/A	N/A
10263476	12/2018	Leabman	N/A	N/A
10270261	12/2018	Bell et al.	N/A	N/A
10277054	12/2018	Hosseini	N/A	N/A
10291055	12/2018	Bell et al.	N/A	N/A
10291056	12/2018	Bell et al.	N/A	N/A
10291066	12/2018	Leabman	N/A	N/A
10291294	12/2018	Leabman	N/A	N/A
10298024	12/2018	Leabman	N/A	N/A
10298133	12/2018	Leabman	N/A	N/A
10305315	12/2018	Leabman et al.	N/A	N/A
10312715	12/2018	Leabman	N/A	N/A
10320446	12/2018	Hosseini	N/A	N/A
10333332	12/2018	Hosseini	N/A	N/A
10355534	12/2018	Johnston et al.	N/A	N/A
10381880	12/2018	Leabman et al.	N/A	N/A
10389161	12/2018	Hosseini et al.	N/A	N/A
10396588	12/2018	Leabman	N/A	N/A
10396604	12/2018	Bell et al.	N/A	N/A
10439442	12/2018	Hosseini et al.	N/A	N/A
10439448	12/2018	Bell et al.	N/A	N/A
10447093	12/2018	Hosseini	N/A	N/A
10476312	12/2018	Johnston et al.	N/A	N/A
10483768	12/2018	Bell et al.	N/A	N/A
10490346	12/2018	Contopanagos	N/A	N/A

10491029	12/2018	Hosseini	N/A	N/A
10498144	12/2018	Leabman et al.	N/A	N/A
10511097	12/2018	Kornaros et al.	N/A	N/A
10511196	12/2018	Hosseini	N/A	N/A
10516289	12/2018	Leabman et al.	N/A	N/A
10516301	12/2018	Leabman	N/A	N/A
10523033	12/2018	Leabman	N/A	N/A
10523058	12/2018	Leabman	N/A	N/A
10554052	12/2019	Bell et al.	N/A	N/A
10594165	12/2019	Hosseini	N/A	N/A
10615647	12/2019	Johnston et al.	N/A	N/A
10680319	12/2019	Hosseini et al.	N/A	N/A
10714984	12/2019	Hosseini et al.	N/A	N/A
10734717	12/2019	Hosseini	N/A	N/A
10778041	12/2019	Leabman	N/A	N/A
10790674	12/2019	Bell et al.	N/A	N/A
10840743	12/2019	Johnston et al.	N/A	N/A
10848853	12/2019	Leabman et al.	N/A	N/A
10879740	12/2019	Hosseini	N/A	N/A
10923954	12/2020	Leabman	N/A	N/A
10958095	12/2020	Leabman et al.	N/A	N/A
10965164	12/2020	Leabman et al.	N/A	N/A
10985617	12/2020	Johnston et al.	N/A	N/A
10992185	12/2020	Leabman	N/A	N/A
10992187	12/2020	Leabman	N/A	N/A
11011942	12/2020	Liu	N/A	N/A
11018779	12/2020	Sarajedini	N/A	N/A
11056929	12/2020	Bell et al.	N/A	N/A
11114885	12/2020	Hosseini et al.	N/A	N/A
11159057	12/2020	Kabiri et al.	N/A	N/A
11218795	12/2021	Leabman et al.	N/A	N/A
11233425	12/2021	Leabman	N/A	N/A
11245191	12/2021	Kornaros et al.	N/A	N/A
11245289	12/2021	Johnston et al.	N/A	N/A
11342798	12/2021	Johnston et al.	N/A	N/A
11355966	12/2021	Muryanto et al.	N/A	N/A
11411437	12/2021	Johnston et al.	N/A	N/A
11437735	12/2021	Papio-Toda et al.	N/A	N/A
11451096	12/2021	Hoss	N/A	N/A
11463179	12/2021	Sarajedini et al.	N/A	N/A
11469629	12/2021	Jain et al.	N/A	N/A
11502551	12/2021	Leabman	N/A	N/A
11515732	12/2021	Contopanagos et al.	N/A	N/A
11539243	12/2021	Katajamaki et al.	N/A	N/A
11594902	12/2022	Johnston et al.	N/A	N/A
11637456	12/2022	Kornaros et al.	N/A	N/A
11652369	12/2022	Leabman et al.	N/A	N/A
11670970	12/2022	Leabman	N/A	N/A
11689045	12/2022	Leabman et al.	N/A	N/A
11699847	12/2022	Contopanagos et al.	N/A	N/A

11710987	12/2022	Johnston et al.	N/A	N/A
2002/0065052	12/2001	Pande et al.	N/A	N/A
2002/0103447	12/2001	Terry	N/A	N/A
2002/0171594	12/2001	Fang	N/A	N/A
2003/0038750	12/2002	Chen	N/A	N/A
2003/0058187	12/2002	Billiet et al.	N/A	N/A
2004/0020100	12/2003	O'Brien et al.	N/A	N/A
2004/0130425	12/2003	Dayan et al.	N/A	N/A
2004/0130442	12/2003	Breed	N/A	N/A
2004/0145342	12/2003	Lyon	N/A	N/A
2004/0155832	12/2003	Yuanzhu	N/A	N/A
2004/0207559	12/2003	Milosavljevic	N/A	N/A
2004/0259604	12/2003	Mickle et al.	N/A	N/A
2005/0007276	12/2004	Barrick et al.	N/A	N/A
2005/0116683	12/2004	Cheng	N/A	N/A
2005/0117660	12/2004	Vialle et al.	N/A	N/A
2005/0134517	12/2004	Gottl	N/A	N/A
2005/0227619	12/2004	Lee et al.	N/A	N/A
2005/0237258	12/2004	Abramov et al.	N/A	N/A
2006/0013335	12/2005	Leabman	N/A	N/A
2006/0019712	12/2005	Choi	N/A	N/A
2006/0030279	12/2005	Leabman et al.	N/A	N/A
2006/0092079	12/2005	de Rochemont	N/A	N/A
2006/0094425	12/2005	Mickle et al.	N/A	N/A
2006/0113955	12/2005	Nunally	N/A	N/A
2006/0119532	12/2005	Yun et al.	N/A	N/A
2006/0160517	12/2005	Yoon	N/A	N/A
2006/0184705	12/2005	Nakajima	N/A	N/A
2006/0199620	12/2005	Greene et al.	N/A	N/A
2006/0238365	12/2005	Vecchione et al.	N/A	N/A
2006/0252370	12/2005	Goossens et al.	N/A	N/A
2006/0266564	12/2005	Perlman et al.	N/A	N/A
2006/0266917	12/2005	Baldis et al.	N/A	N/A
2006/0284593	12/2005	Nagy et al.	N/A	N/A
2007/0007821	12/2006	Rossetti	N/A	N/A
2007/0019693	12/2006	Graham	N/A	N/A
2007/0021140	12/2006	Keyes, IV et al.	N/A	N/A
2007/0060185	12/2006	Simon et al.	N/A	N/A
2007/0090997	12/2006	Brown et al.	N/A	N/A
2007/0093269	12/2006	Leabman et al.	N/A	N/A
2007/0097653	12/2006	Gilliland et al.	N/A	N/A
2007/0103110	12/2006	Sagoo	N/A	N/A
2007/0106894	12/2006	Zhang	N/A	N/A
2007/0109121	12/2006	Cohen	N/A	N/A
2007/0139000	12/2006	Kozuma	N/A	N/A
2007/0149162	12/2006	Greene et al.	N/A	N/A
2007/0164868	12/2006	Deavours et al.	N/A	N/A
2007/0173214	12/2006	Mickle et al.	N/A	N/A
2007/0178857	12/2006	Greene et al.	N/A	N/A
2007/0178945	12/2006	Cook et al.	N/A	N/A

2007/0182367	12/2006	Partovi	N/A	N/A
2007/0191074	12/2006	Harrist et al.	N/A	N/A
2007/0191075	12/2006	Greene et al.	N/A	N/A
2007/0210960	12/2006	Rofougaran et al.	N/A	N/A
2007/0222681	12/2006	Greene et al.	N/A	N/A
2007/0228833	12/2006	Stevens et al.	N/A	N/A
2007/0229261	12/2006	Zimmerman et al.	N/A	N/A
2007/0240297	12/2006	Yang et al.	N/A	N/A
2007/0273486	12/2006	Shiotsu	N/A	N/A
2007/0296639	12/2006	Hook et al.	N/A	N/A
2007/0298846	12/2006	Greene et al.	N/A	N/A
2008/0014897	12/2007	Cook et al.	N/A	N/A
2008/0024376	12/2007	Norris et al.	N/A	N/A
2008/0048917	12/2007	Achour et al.	N/A	N/A
2008/0067874	12/2007	Tseng	N/A	N/A
2008/0074324	12/2007	Puzella et al.	N/A	N/A
2008/0089277	12/2007	Alexander et al.	N/A	N/A
2008/0110263	12/2007	Klessel et al.	N/A	N/A
2008/0122297	12/2007	Arai	N/A	N/A
2008/0123383	12/2007	Shionoiri	N/A	N/A
2008/0169910	12/2007	Greene et al.	N/A	N/A
2008/0197802	12/2007	Onishi	N/A	N/A
2008/0204350	12/2007	Tam et al.	N/A	N/A
2008/0210762	12/2007	Osada et al.	N/A	N/A
2008/0211458	12/2007	Lawther et al.	N/A	N/A
2008/0233890	12/2007	Baker	N/A	N/A
2008/0258993	12/2007	Gummalla et al.	N/A	N/A
2008/0266191	12/2007	Hilgers	N/A	N/A
2008/0278378	12/2007	Chang et al.	N/A	N/A
2008/0309452	12/2007	Zeine	N/A	N/A
2009/0002493	12/2008	Kates	N/A	N/A
2009/0010316	12/2008	Rofougaran et al.	N/A	N/A
2009/0019183	12/2008	Wu et al.	N/A	N/A
2009/0036065	12/2008	Siu	N/A	N/A
2009/0039828	12/2008	Jakubowski	N/A	N/A
2009/0047998	12/2008	Alberth, Jr.	N/A	N/A
2009/0058361	12/2008	John	N/A	N/A
2009/0058731	12/2008	Geary et al.	N/A	N/A
2009/0060012	12/2008	Gresset et al.	N/A	N/A
2009/0067198	12/2008	Graham et al.	N/A	N/A
2009/0067208	12/2008	Martin et al.	N/A	N/A
2009/0073066	12/2008	Jordon et al.	N/A	N/A
2009/0096412	12/2008	Huang	N/A	N/A
2009/0096413	12/2008	Partovi	N/A	N/A
2009/0102292	12/2008	Cook et al.	N/A	N/A
2009/0102296	12/2008	Greene et al.	N/A	N/A
2009/0108679	12/2008	Porwal	N/A	N/A
2009/0122847	12/2008	Nysen et al.	N/A	N/A
2009/0128262	12/2008	Lee et al.	N/A	N/A
2009/0174604	12/2008	Keskitalo	N/A	N/A

2009/0200985	12/2008	Zane et al.	N/A	N/A
2009/0206791	12/2008	Jung	N/A	N/A
2009/0207092	12/2008	Nysen et al.	N/A	N/A
2009/0218884	12/2008	Soar	N/A	N/A
2009/0218891	12/2008	McCollough	N/A	N/A
2009/0243397	12/2008	Cook et al.	N/A	N/A
2009/0256752	12/2008	Akkermans et al.	N/A	N/A
2009/0264069	12/2008	Yamasuge	N/A	N/A
2009/0271048	12/2008	Wakamatsu	N/A	N/A
2009/0281678	12/2008	Wakamatsu	N/A	N/A
2009/0284082	12/2008	Mohammadian	N/A	N/A
2009/0284220	12/2008	Toncich et al.	N/A	N/A
2009/0284227	12/2008	Mohammadian et al.	N/A	N/A
2009/0286475	12/2008	Toncich et al.	N/A	N/A
2009/0286476	12/2008	Toncich et al.	N/A	N/A
2009/0291634	12/2008	Saarisalo	N/A	N/A
2009/0312046	12/2008	Clevenger et al.	N/A	N/A
2009/0322281	12/2008	Kamijo et al.	N/A	N/A
2010/0001683	12/2009	Huang et al.	N/A	N/A
2010/0007307	12/2009	Baarmaan et al.	N/A	N/A
2010/0007569	12/2009	Sim et al.	N/A	N/A
2010/0019908	12/2009	Cho et al.	N/A	N/A
2010/0033021	12/2009	Bennett	N/A	N/A
2010/0034238	12/2009	Bennett	N/A	N/A
2010/0044123	12/2009	Perlman et al.	N/A	N/A
2010/0060534	12/2009	Oodachi	N/A	N/A
2010/0066631	12/2009	Puzella et al.	N/A	N/A
2010/0075607	12/2009	Hosoya	N/A	N/A
2010/0079005	12/2009	Hyde et al.	N/A	N/A
2010/0079011	12/2009	Hyde et al.	N/A	N/A
2010/0087227	12/2009	Francos et al.	N/A	N/A
2010/0090656	12/2009	Shearer et al.	N/A	N/A
2010/0142418	12/2009	Nishioka et al.	N/A	N/A
2010/0142509	12/2009	Zhu et al.	N/A	N/A
2010/0148723	12/2009	Cook et al.	N/A	N/A
2010/0151808	12/2009	Toncich et al.	N/A	N/A
2010/0156741	12/2009	Vazquez et al.	N/A	N/A
2010/0164296	12/2009	Kurs et al.	N/A	N/A
2010/0164433	12/2009	Janefalker et al.	N/A	N/A
2010/0167664	12/2009	Szini	N/A	N/A
2010/0171461	12/2009	Baarmaan et al.	N/A	N/A
2010/0171676	12/2009	Tani et al.	N/A	N/A
2010/0174629	12/2009	Taylor et al.	N/A	N/A
2010/0176934	12/2009	Chou et al.	N/A	N/A
2010/0181961	12/2009	Novak et al.	N/A	N/A
2010/0181964	12/2009	Huggins et al.	N/A	N/A
2010/0194206	12/2009	Burdo et al.	N/A	N/A
2010/0201189	12/2009	Kirby et al.	N/A	N/A
2010/0201201	12/2009	Mobarhan et al.	N/A	N/A
2010/0201314	12/2009	Toncich et al.	N/A	N/A

2010/0207572	12/2009	Kirby et al.	N/A	N/A
2010/0210233	12/2009	Cook et al.	N/A	N/A
2010/0213895	12/2009	Keating et al.	N/A	N/A
2010/0214177	12/2009	Parsche	N/A	N/A
2010/0222010	12/2009	Ozaki et al.	N/A	N/A
2010/0225270	12/2009	Jacobs et al.	N/A	N/A
2010/0227570	12/2009	Hendin	N/A	N/A
2010/0244576	12/2009	Hillan et al.	N/A	N/A
2010/0253281	12/2009	Li	N/A	N/A
2010/0256831	12/2009	Abramo et al.	N/A	N/A
2010/0264747	12/2009	Hall et al.	N/A	N/A
2010/0277003	12/2009	Von Novak et al.	N/A	N/A
2010/0279606	12/2009	Hillan et al.	N/A	N/A
2010/0289341	12/2009	Ozaki et al.	N/A	N/A
2010/0295372	12/2009	Hyde et al.	N/A	N/A
2010/0309088	12/2009	Hyvonen et al.	N/A	N/A
2010/0315045	12/2009	Zeine	N/A	N/A
2010/0328044	12/2009	Waffenschmidt et al.	N/A	N/A
2011/0009057	12/2010	Saunamäki	N/A	N/A
2011/0013198	12/2010	Shirley	N/A	N/A
2011/0018360	12/2010	Baarma et al.	N/A	N/A
2011/0028114	12/2010	Kerselaers	N/A	N/A
2011/0032149	12/2010	Leabman	N/A	N/A
2011/0032866	12/2010	Leabman	N/A	N/A
2011/0034190	12/2010	Leabman	N/A	N/A
2011/0034191	12/2010	Leabman	N/A	N/A
2011/0050166	12/2010	Cook et al.	N/A	N/A
2011/0057607	12/2010	Carobolante	N/A	N/A
2011/0057853	12/2010	Kim et al.	N/A	N/A
2011/0062788	12/2010	Chen et al.	N/A	N/A
2011/0074342	12/2010	MacLaughlin	N/A	N/A
2011/0074349	12/2010	Ghovanloo	N/A	N/A
2011/0092161	12/2010	Dotan	N/A	N/A
2011/0109167	12/2010	Park et al.	N/A	N/A
2011/0115303	12/2010	Baarma et al.	N/A	N/A
2011/0115432	12/2010	El-Maleh	N/A	N/A
2011/0115605	12/2010	Dimig et al.	N/A	N/A
2011/0121660	12/2010	Azancot et al.	N/A	N/A
2011/0122018	12/2010	Tarng et al.	N/A	N/A
2011/0122026	12/2010	DeLaquil et al.	N/A	N/A
2011/0127845	12/2010	Walley et al.	N/A	N/A
2011/0127952	12/2010	Walley et al.	N/A	N/A
2011/0133691	12/2010	Hautanen	N/A	N/A
2011/0156493	12/2010	Bennett	N/A	N/A
2011/0156494	12/2010	Mashinsky	N/A	N/A
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2011/0175455	12/2010	Hashiguchi	N/A	N/A
2011/0175461	12/2010	Tinaphong	N/A	N/A
2011/0181120	12/2010	Liu et al.	N/A	N/A
2011/0182245	12/2010	Malkamaki et al.	N/A	N/A

2011/0184842	12/2010	Melen	N/A	N/A
2011/0194543	12/2010	Zhao et al.	N/A	N/A
2011/0195722	12/2010	Walter et al.	N/A	N/A
2011/0199046	12/2010	Tsai et al.	N/A	N/A
2011/0202289	12/2010	Kalantari Khandani	N/A	N/A
2011/0215086	12/2010	Yeh	N/A	N/A
2011/0217923	12/2010	Ma	N/A	N/A
2011/0220634	12/2010	Yeh	N/A	N/A
2011/0221389	12/2010	Won et al.	N/A	N/A
2011/0222154	12/2010	Choi et al.	N/A	N/A
2011/0222272	12/2010	Yeh	N/A	N/A
2011/0227725	12/2010	Muirhead	N/A	N/A
2011/0243040	12/2010	Khan et al.	N/A	N/A
2011/0243050	12/2010	Yanover	N/A	N/A
2011/0244913	12/2010	Kim et al.	N/A	N/A
2011/0248573	12/2010	Kanno et al.	N/A	N/A
2011/0248575	12/2010	Kim et al.	N/A	N/A
2011/0249678	12/2010	Bonicatto	N/A	N/A
2011/0254377	12/2010	Wildmer et al.	N/A	N/A
2011/0254503	12/2010	Widmer et al.	N/A	N/A
2011/0259953	12/2010	Baarman et al.	N/A	N/A
2011/0273977	12/2010	Shapira et al.	N/A	N/A
2011/0278941	12/2010	Krishna et al.	N/A	N/A
2011/0279226	12/2010	Chen et al.	N/A	N/A
2011/0281535	12/2010	Low et al.	N/A	N/A
2011/0282415	12/2010	Eckhoff et al.	N/A	N/A
2011/0285213	12/2010	Kowalewski	N/A	N/A
2011/0286374	12/2010	Shin et al.	N/A	N/A
2011/0291489	12/2010	Tsai et al.	N/A	N/A
2011/0302078	12/2010	Failing	N/A	N/A
2011/0304216	12/2010	Baarman	N/A	N/A
2011/0304437	12/2010	Beeler	N/A	N/A
2011/0304521	12/2010	Ando et al.	N/A	N/A
2012/0007441	12/2011	John	N/A	N/A
2012/0013196	12/2011	Kim et al.	N/A	N/A
2012/0013198	12/2011	Uramoto et al.	N/A	N/A
2012/0013296	12/2011	Heydari et al.	N/A	N/A
2012/0019419	12/2011	Prat et al.	N/A	N/A
2012/0025622	12/2011	Kim et al.	N/A	N/A
2012/0043887	12/2011	Mesibov	N/A	N/A
2012/0051109	12/2011	Kim et al.	N/A	N/A
2012/0051294	12/2011	Guillouard	N/A	N/A
2012/0056486	12/2011	Endo et al.	N/A	N/A
2012/0056741	12/2011	Zhu et al.	N/A	N/A
2012/0068906	12/2011	Asher et al.	N/A	N/A
2012/0074891	12/2011	Anderson et al.	N/A	N/A
2012/0075072	12/2011	Pappu	N/A	N/A
2012/0080944	12/2011	Recker et al.	N/A	N/A
2012/0080957	12/2011	Cooper et al.	N/A	N/A
2012/0086284	12/2011	Capanella et al.	N/A	N/A

2012/0086615	12/2011	Norair	N/A	N/A
2012/0095617	12/2011	Martin et al.	N/A	N/A
2012/0098350	12/2011	Campanella et al.	N/A	N/A
2012/0098485	12/2011	Kang et al.	N/A	N/A
2012/0099675	12/2011	Kitamura et al.	N/A	N/A
2012/0103562	12/2011	Alexander	N/A	N/A
2012/0104849	12/2011	Jackson	N/A	N/A
2012/0105252	12/2011	Wang	N/A	N/A
2012/0112532	12/2011	Kesler et al.	N/A	N/A
2012/0119914	12/2011	Uchida	N/A	N/A
2012/0126743	12/2011	Rivers, Jr.	N/A	N/A
2012/0132647	12/2011	Beverly et al.	N/A	N/A
2012/0133214	12/2011	Yun et al.	N/A	N/A
2012/0142291	12/2011	Rath et al.	N/A	N/A
2012/0146426	12/2011	Sabo	N/A	N/A
2012/0146576	12/2011	Partovi	N/A	N/A
2012/0146577	12/2011	Tanabe	N/A	N/A
2012/0147802	12/2011	Ukita et al.	N/A	N/A
2012/0149307	12/2011	Terada et al.	N/A	N/A
2012/0150670	12/2011	Taylor et al.	N/A	N/A
2012/0153731	12/2011	Kirby	307/104	H02J 50/20
2012/0153894	12/2011	Widmer et al.	N/A	N/A
2012/0157019	12/2011	Li	N/A	N/A
2012/0161531	12/2011	Kim et al.	N/A	N/A
2012/0161544	12/2011	Kashiwagi et al.	N/A	N/A
2012/0169276	12/2011	Wang	N/A	N/A
2012/0169278	12/2011	Choi	N/A	N/A
2012/0173418	12/2011	Beardsmore et al.	N/A	N/A
2012/0179004	12/2011	Roesicke et al.	N/A	N/A
2012/0181973	12/2011	Lyden	N/A	N/A
2012/0182427	12/2011	Marshall	N/A	N/A
2012/0187851	12/2011	Huggins et al.	N/A	N/A
2012/0188142	12/2011	Shashi et al.	N/A	N/A
2012/0193999	12/2011	Zeine	N/A	N/A
2012/0200399	12/2011	Chae	N/A	N/A
2012/0201153	12/2011	Bharadia et al.	N/A	N/A
2012/0201173	12/2011	Jain et al.	N/A	N/A
2012/0206299	12/2011	Valdes-Garcia	N/A	N/A
2012/0211214	12/2011	Phan	N/A	N/A
2012/0212071	12/2011	Miyabayashi et al.	N/A	N/A
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2012/0214462	12/2011	Chu et al.	N/A	N/A
2012/0214536	12/2011	Kim et al.	N/A	N/A
2012/0228392	12/2011	Cameron et al.	N/A	N/A
2012/0228956	12/2011	Kamata	N/A	N/A
2012/0231856	12/2011	Lee et al.	N/A	N/A
2012/0235636	12/2011	Partovi	N/A	N/A
2012/0242283	12/2011	Kim et al.	N/A	N/A
2012/0248886	12/2011	Kesler et al.	N/A	N/A
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2012/0248891	12/2011	Drennen	N/A	N/A
2012/0249051	12/2011	Son et al.	N/A	N/A
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2012/0267900	12/2011	Huffman et al.	N/A	N/A
2012/0268238	12/2011	Park et al.	N/A	N/A
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2012/0274154	12/2011	DeLuca	N/A	N/A
2012/0280650	12/2011	Kim et al.	N/A	N/A
2012/0286582	12/2011	Kim et al.	N/A	N/A
2012/0292993	12/2011	Mettler et al.	N/A	N/A
2012/0293021	12/2011	Teggatz et al.	N/A	N/A
2012/0293119	12/2011	Park et al.	N/A	N/A
2012/0299389	12/2011	Lee et al.	N/A	N/A
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2012/0306284	12/2011	Lee et al.	N/A	N/A
2012/0306433	12/2011	Kim et al.	N/A	N/A
2012/0306572	12/2011	Hietala et al.	N/A	N/A
2012/0306705	12/2011	Sakurai et al.	N/A	N/A
2012/0306707	12/2011	Yang et al.	N/A	N/A
2012/0306720	12/2011	Tamai et al.	N/A	N/A
2012/0307873	12/2011	Kim et al.	N/A	N/A
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2012/0309308	12/2011	Kim et al.	N/A	N/A
2012/0309332	12/2011	Liao	N/A	N/A
2012/0313446	12/2011	Park et al.	N/A	N/A
2012/0313449	12/2011	Kurs	N/A	N/A
2012/0313835	12/2011	Gebretnsae	N/A	N/A
2012/0326660	12/2011	Lu et al.	N/A	N/A
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2013/0005252	12/2012	Lee et al.	N/A	N/A
2013/0018439	12/2012	Chow et al.	N/A	N/A
2013/0024059	12/2012	Miller et al.	N/A	N/A
2013/0026981	12/2012	Van Der Lee	N/A	N/A
2013/0026982	12/2012	Rothenbaum	N/A	N/A
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2013/0033571	12/2012	Steen	N/A	N/A
2013/0038124	12/2012	Newdoll et al.	N/A	N/A
2013/0038402	12/2012	Karalis et al.	N/A	N/A
2013/0043738	12/2012	Park et al.	N/A	N/A
2013/0044035	12/2012	Zhuang	N/A	N/A
2013/0049471	12/2012	Oleynik	N/A	N/A
2013/0049475	12/2012	Kim et al.	N/A	N/A
2013/0049484	12/2012	Weissentern et al.	N/A	N/A
2013/0057078	12/2012	Lee	N/A	N/A

2013/0057205	12/2012	Lee et al.	N/A	N/A
2013/0057210	12/2012	Nergaard et al.	N/A	N/A
2013/0057364	12/2012	Kesler et al.	N/A	N/A
2013/0058379	12/2012	Kim et al.	N/A	N/A
2013/0062959	12/2012	Lee et al.	N/A	N/A
2013/0063082	12/2012	Lee et al.	N/A	N/A
2013/0063143	12/2012	Adalsteinsson et al.	N/A	N/A
2013/0063266	12/2012	Yunker et al.	N/A	N/A
2013/0069444	12/2012	Waffenschmidt et al.	N/A	N/A
2013/0076308	12/2012	Niskala et al.	N/A	N/A
2013/0077650	12/2012	Traxler et al.	N/A	N/A
2013/0078918	12/2012	Crowley et al.	N/A	N/A
2013/0082651	12/2012	Park et al.	N/A	N/A
2013/0082653	12/2012	Lee et al.	N/A	N/A
2013/0083774	12/2012	Son et al.	N/A	N/A
2013/0088082	12/2012	Kang et al.	N/A	N/A
2013/0088090	12/2012	Wu	N/A	N/A
2013/0088192	12/2012	Eaton	N/A	N/A
2013/0088331	12/2012	Cho	N/A	N/A
2013/0093388	12/2012	Partovi	N/A	N/A
2013/0099389	12/2012	Hong et al.	N/A	N/A
2013/0099586	12/2012	Kato	N/A	N/A
2013/0106197	12/2012	Bae et al.	N/A	N/A
2013/0107023	12/2012	Tanaka et al.	N/A	N/A
2013/0119777	12/2012	Rees	N/A	N/A
2013/0119778	12/2012	Jung	N/A	N/A
2013/0119929	12/2012	Partovi	N/A	N/A
2013/0120052	12/2012	Siska	N/A	N/A
2013/0120205	12/2012	Thomson et al.	N/A	N/A
2013/0120206	12/2012	Biancotto et al.	N/A	N/A
2013/0120217	12/2012	Ueda et al.	N/A	N/A
2013/0130621	12/2012	Kim et al.	N/A	N/A
2013/0132010	12/2012	Winger et al.	N/A	N/A
2013/0134923	12/2012	Smith	N/A	N/A
2013/0137455	12/2012	Xia	N/A	N/A
2013/0141037	12/2012	Jenwatanavet et al.	N/A	N/A
2013/0148341	12/2012	Williams	N/A	N/A
2013/0149975	12/2012	Yu et al.	N/A	N/A
2013/0154387	12/2012	Lee et al.	N/A	N/A
2013/0155748	12/2012	Sundstrom	N/A	N/A
2013/0157729	12/2012	Tabe	N/A	N/A
2013/0162335	12/2012	Kim et al.	N/A	N/A
2013/0169061	12/2012	Miroshnichenko et al.	N/A	N/A
2013/0169219	12/2012	Gray	N/A	N/A
2013/0169348	12/2012	Shi	N/A	N/A
2013/0171939	12/2012	Tian et al.	N/A	N/A
2013/0175877	12/2012	Abe et al.	N/A	N/A
2013/0178253	12/2012	Karaoguz	N/A	N/A
2013/0181881	12/2012	Christie et al.	N/A	N/A

2013/0187475	12/2012	Vendik	N/A	N/A
2013/0190031	12/2012	Persson et al.	N/A	N/A
2013/0193769	12/2012	Mehta et al.	N/A	N/A
2013/0197320	12/2012	Albert et al.	N/A	N/A
2013/0200064	12/2012	Alexander	N/A	N/A
2013/0207477	12/2012	Nam et al.	N/A	N/A
2013/0207604	12/2012	Zeine	N/A	N/A
2013/0207879	12/2012	Rada et al.	N/A	N/A
2013/0207893	12/2012	Liu et al.	N/A	N/A
2013/0210357	12/2012	Qin et al.	N/A	N/A
2013/0221757	12/2012	Cho et al.	N/A	N/A
2013/0222201	12/2012	Ma et al.	N/A	N/A
2013/0234530	12/2012	Miyauchi	N/A	N/A
2013/0234536	12/2012	Chemishkian et al.	N/A	N/A
2013/0234658	12/2012	Endo et al.	N/A	N/A
2013/0241306	12/2012	Aber et al.	N/A	N/A
2013/0241468	12/2012	Moshfeghi	N/A	N/A
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2013/0249682	12/2012	Van Wiemeersch et al.	N/A	N/A
2013/0250102	12/2012	Scanlon et al.	N/A	N/A
2013/0254578	12/2012	Huang et al.	N/A	N/A
2013/0264997	12/2012	Lee et al.	N/A	N/A
2013/0268782	12/2012	Tam et al.	N/A	N/A
2013/0270923	12/2012	Cook et al.	N/A	N/A
2013/0278076	12/2012	Proud	N/A	N/A
2013/0278209	12/2012	Von Novak, III et al.	N/A	N/A
2013/0285464	12/2012	Miwa	N/A	N/A
2013/0285477	12/2012	Lo et al.	N/A	N/A
2013/0285606	12/2012	Ben-Shalom et al.	N/A	N/A
2013/0288600	12/2012	Kuusilinna et al.	N/A	N/A
2013/0288617	12/2012	Kim et al.	N/A	N/A
2013/0293423	12/2012	Moshfeghi	N/A	N/A
2013/0300356	12/2012	Yang	N/A	N/A
2013/0307751	12/2012	Yu-Juin et al.	N/A	N/A
2013/0310020	12/2012	Kazuhiro	N/A	N/A
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2013/0328417	12/2012	Takeuchi	N/A	N/A
2013/0334883	12/2012	Kim et al.	N/A	N/A
2013/0339108	12/2012	Ryder et al.	N/A	N/A
2013/0343208	12/2012	Sexton et al.	N/A	N/A
2013/0343251	12/2012	Zhang	N/A	N/A
2013/0343585	12/2012	Bennett et al.	N/A	N/A
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2014/0001875	12/2013	Nahidipour	N/A	N/A
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2014/0015336	12/2013	Weber et al.	N/A	N/A
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2014/0021907	12/2013	Yu et al.	N/A	N/A
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2014/0057618	12/2013	Zirwas et al.	N/A	N/A
2014/0062395	12/2013	Kwon et al.	N/A	N/A
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2014/0086592	12/2013	Nakahara et al.	N/A	N/A
2014/0091756	12/2013	Ofstein et al.	N/A	N/A
2014/0091968	12/2013	Harel et al.	N/A	N/A
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2014/0139039	12/2013	Cook et al.	N/A	N/A
2014/0139180	12/2013	Kim et al.	N/A	N/A
2014/0141838	12/2013	Cai et al.	N/A	N/A
2014/0142876	12/2013	John et al.	N/A	N/A
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2014/0159667	12/2013	Kim et al.	N/A	N/A
2014/0169385	12/2013	Hadani et al.	N/A	N/A
2014/0175893	12/2013	Sengupta et al.	N/A	N/A
2014/0176054	12/2013	Porat et al.	N/A	N/A
2014/0176061	12/2013	Cheatham, III et al.	N/A	N/A
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2014/0177399	12/2013	Teng et al.	N/A	N/A
2014/0183964	12/2013	Walley	N/A	N/A
2014/0184148	12/2013	Van Der Lee et al.	N/A	N/A
2014/0184155	12/2013	Cha	N/A	N/A
2014/0184163	12/2013	Das et al.	N/A	N/A
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2014/0194092	12/2013	Wanstedt et al.	N/A	N/A
2014/0194095	12/2013	Wynne et al.	N/A	N/A
2014/0197691	12/2013	Wang	N/A	N/A
2014/0203629	12/2013	Hoffman et al.	N/A	N/A
2014/0206384	12/2013	Kim et al.	N/A	N/A
2014/0210281	12/2013	Ito et al.	N/A	N/A
2014/0217955	12/2013	Lin	N/A	N/A
2014/0217967	12/2013	Zeine et al.	N/A	N/A
2014/0225805	12/2013	Pan et al.	N/A	N/A
2014/0232320	12/2013	Ento July et al.	N/A	N/A
2014/0232610	12/2013	Shigemoto et al.	N/A	N/A
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2014/0252813	12/2013	Lee et al.	N/A	N/A
2014/0252866	12/2013	Walsh et al.	N/A	N/A
2014/0265725	12/2013	Angle et al.	N/A	N/A
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2014/0265943	12/2013	Angle et al.	N/A	N/A
2014/0266025	12/2013	Jakubowski	N/A	N/A
2014/0266946	12/2013	Bily et al.	N/A	N/A
2014/0273819	12/2013	Nadakuduti et al.	N/A	N/A
2014/0273892	12/2013	Nourbakhsh	N/A	N/A
2014/0281655	12/2013	Angle et al.	N/A	N/A
2014/0292090	12/2013	Cordeiro et al.	N/A	N/A
2014/0292451	12/2013	Zimmerman	N/A	N/A
2014/0300452	12/2013	Rofe et al.	N/A	N/A
2014/0312706	12/2013	Fiorello et al.	N/A	N/A
2014/0325218	12/2013	Shimizu et al.	N/A	N/A
2014/0327320	12/2013	Muhs et al.	N/A	N/A
2014/0327390	12/2013	Park et al.	N/A	N/A
2014/0333142	12/2013	Desrosiers	N/A	N/A
2014/0346860	12/2013	Aubry et al.	N/A	N/A
2014/0354063	12/2013	Leabman et al.	N/A	N/A
2014/0354221	12/2013	Leabman et al.	N/A	N/A
2014/0355718	12/2013	Guan et al.	N/A	N/A
2014/0368048	12/2013	Leabman et al.	N/A	N/A
2014/0368161	12/2013	Leabman et al.	N/A	N/A
2014/0368405	12/2013	Ek et al.	N/A	N/A
2014/0375139	12/2013	Tsukamoto	N/A	N/A

2014/0375253	12/2013	Leabman et al.	N/A	N/A
2014/0375258	12/2013	Arkhipenkov	N/A	N/A
2014/0375261	12/2013	Manova-Elssibony et al.	N/A	N/A
2015/0001949	12/2014	Leabman et al.	N/A	N/A
2015/0002080	12/2014	Lang	N/A	N/A
2015/0002086	12/2014	Matos et al.	N/A	N/A
2015/0003207	12/2014	Lee et al.	N/A	N/A
2015/0008980	12/2014	Kim et al.	N/A	N/A
2015/0011160	12/2014	Jurgovan et al.	N/A	N/A
2015/0015180	12/2014	Miller et al.	N/A	N/A
2015/0015182	12/2014	Brandtman et al.	N/A	N/A
2015/0015192	12/2014	Leabman et al.	N/A	N/A
2015/0021990	12/2014	Myer et al.	N/A	N/A
2015/0022008	12/2014	Leabman et al.	N/A	N/A
2015/0022010	12/2014	Leabman et al.	N/A	N/A
2015/0022194	12/2014	Almalki et al.	N/A	N/A
2015/0023204	12/2014	Wik et al.	N/A	N/A
2015/0028688	12/2014	Masaoka	N/A	N/A
2015/0028694	12/2014	Leabman et al.	N/A	N/A
2015/0028697	12/2014	Makino et al.	N/A	N/A
2015/0028875	12/2014	Irie et al.	N/A	N/A
2015/0035378	12/2014	Calhoun et al.	N/A	N/A
2015/0035709	12/2014	Lim	N/A	N/A
2015/0035715	12/2014	Kim et al.	N/A	N/A
2015/0039482	12/2014	Fujinaga	N/A	N/A
2015/0041459	12/2014	Leabman et al.	N/A	N/A
2015/0042265	12/2014	Leabman et al.	N/A	N/A
2015/0044977	12/2014	Ramasamy et al.	N/A	N/A
2015/0046526	12/2014	Bush et al.	N/A	N/A
2015/0061404	12/2014	Lamenza et al.	N/A	N/A
2015/0076917	12/2014	Leabman et al.	N/A	N/A
2015/0076927	12/2014	Leabman et al.	N/A	N/A
2015/0077036	12/2014	Leabman et al.	N/A	N/A
2015/0077037	12/2014	Leabman et al.	N/A	N/A
2015/0091520	12/2014	Blum et al.	N/A	N/A
2015/0091706	12/2014	Chemishkian et al.	N/A	N/A
2015/0097442	12/2014	Muurinen	N/A	N/A
2015/0097663	12/2014	Sloo et al.	N/A	N/A
2015/0102764	12/2014	Leabman et al.	N/A	N/A
2015/0102769	12/2014	Leabman et al.	N/A	N/A
2015/0102942	12/2014	Houser et al.	N/A	N/A
2015/0102973	12/2014	Hand et al.	N/A	N/A
2015/0108848	12/2014	Joehren	N/A	N/A
2015/0109181	12/2014	Hyde et al.	N/A	N/A
2015/0115877	12/2014	Arai et al.	N/A	N/A
2015/0115878	12/2014	Park	N/A	N/A
2015/0116153	12/2014	Chen et al.	N/A	N/A
2015/0128733	12/2014	Taylor et al.	N/A	N/A
2015/0130285	12/2014	Leabman et al.	N/A	N/A

2015/0130293	12/2014	Hajimiri et al.	N/A	N/A
2015/0137612	12/2014	Yamakawa et al.	N/A	N/A
2015/0148664	12/2014	Stolka et al.	N/A	N/A
2015/0155737	12/2014	Mayo	N/A	N/A
2015/0155738	12/2014	Leabman et al.	N/A	N/A
2015/0162129	12/2014	Fujieda	N/A	N/A
2015/0162662	12/2014	Chen et al.	N/A	N/A
2015/0162751	12/2014	Leabman et al.	N/A	N/A
2015/0162779	12/2014	Lee et al.	N/A	N/A
2015/0171512	12/2014	Chen et al.	N/A	N/A
2015/0171513	12/2014	Chen et al.	N/A	N/A
2015/0171656	12/2014	Leabman et al.	N/A	N/A
2015/0171658	12/2014	Manova-Elssibony et al.	N/A	N/A
2015/0171931	12/2014	Won et al.	N/A	N/A
2015/0177326	12/2014	Chakraborty et al.	N/A	N/A
2015/0180133	12/2014	Hunt	N/A	N/A
2015/0180249	12/2014	Jeon et al.	N/A	N/A
2015/0180284	12/2014	Kang et al.	N/A	N/A
2015/0181117	12/2014	Park et al.	N/A	N/A
2015/0187491	12/2014	Yanagawa	N/A	N/A
2015/0188352	12/2014	Peek et al.	N/A	N/A
2015/0199665	12/2014	Chu	N/A	N/A
2015/0201385	12/2014	Mercer et al.	N/A	N/A
2015/0207333	12/2014	Baerman et al.	N/A	N/A
2015/0207542	12/2014	Zeine	N/A	N/A
2015/0222126	12/2014	Leabman et al.	N/A	N/A
2015/0233987	12/2014	Von Novak, III et al.	N/A	N/A
2015/0234144	12/2014	Cameron et al.	N/A	N/A
2015/0236515	12/2014	Yoshida et al.	N/A	N/A
2015/0236520	12/2014	Baerman	N/A	N/A
2015/0236877	12/2014	Peng et al.	N/A	N/A
2015/0244070	12/2014	Cheng et al.	N/A	N/A
2015/0244080	12/2014	Gregoire	N/A	N/A
2015/0244187	12/2014	Horie	N/A	N/A
2015/0244201	12/2014	Chu	N/A	N/A
2015/0244341	12/2014	Ritter et al.	N/A	N/A
2015/0249484	12/2014	Mach et al.	N/A	N/A
2015/0255989	12/2014	Walley et al.	N/A	N/A
2015/0256097	12/2014	Gudan et al.	N/A	N/A
2015/0260835	12/2014	Widmer et al.	N/A	N/A
2015/0262465	12/2014	Pritchett	N/A	N/A
2015/0263534	12/2014	Lee et al.	N/A	N/A
2015/0263548	12/2014	Cooper	N/A	N/A
2015/0270618	12/2014	Zhu et al.	N/A	N/A
2015/0270622	12/2014	Takasaki et al.	N/A	N/A
2015/0270741	12/2014	Leabman et al.	N/A	N/A
2015/0278558	12/2014	Priev et al.	N/A	N/A
2015/0280429	12/2014	Makita et al.	N/A	N/A
2015/0280484	12/2014	Radziemski et al.	N/A	N/A

2015/0288067	12/2014	Kwon et al.	N/A	N/A
2015/0288074	12/2014	Harper et al.	N/A	N/A
2015/0288438	12/2014	Maltsev et al.	N/A	N/A
2015/0311585	12/2014	Church et al.	N/A	N/A
2015/0312721	12/2014	Singh	N/A	N/A
2015/0318729	12/2014	Leabman	N/A	N/A
2015/0326024	12/2014	Bell et al.	N/A	N/A
2015/0326070	12/2014	Petras et al.	N/A	N/A
2015/0326072	12/2014	Petras et al.	N/A	N/A
2015/0326143	12/2014	Petras et al.	N/A	N/A
2015/0327085	12/2014	Hadani	N/A	N/A
2015/0333528	12/2014	Leabman	N/A	N/A
2015/0333573	12/2014	Leabman	N/A	N/A
2015/0333800	12/2014	Perry et al.	N/A	N/A
2015/0339497	12/2014	Kurian	N/A	N/A
2015/0340759	12/2014	Bridgelall et al.	N/A	N/A
2015/0340872	12/2014	Franzon	307/104	H02J 50/402
2015/0340903	12/2014	Bell et al.	N/A	N/A
2015/0341087	12/2014	Moore et al.	N/A	N/A
2015/0358222	12/2014	Berger et al.	N/A	N/A
2015/0365137	12/2014	Miller et al.	N/A	N/A
2015/0365138	12/2014	Miller et al.	N/A	N/A
2016/0005068	12/2015	Im et al.	N/A	N/A
2016/0012695	12/2015	Bell et al.	N/A	N/A
2016/0013560	12/2015	Daniels	N/A	N/A
2016/0013661	12/2015	Kurs et al.	N/A	N/A
2016/0013677	12/2015	Bell et al.	N/A	N/A
2016/0013855	12/2015	Campos	N/A	N/A
2016/0020636	12/2015	Khlat	N/A	N/A
2016/0028403	12/2015	McCaughan et al.	N/A	N/A
2016/0033254	12/2015	Zeine et al.	N/A	N/A
2016/0042206	12/2015	Pesavento et al.	N/A	N/A
2016/0043571	12/2015	Kesler et al.	N/A	N/A
2016/0043572	12/2015	Cooper et al.	N/A	N/A
2016/0054440	12/2015	Younis	N/A	N/A
2016/0056635	12/2015	Bell	N/A	N/A
2016/0056640	12/2015	Mao	N/A	N/A
2016/0065005	12/2015	Won et al.	N/A	N/A
2016/0079799	12/2015	Khlat	N/A	N/A
2016/0087483	12/2015	Hietala et al.	N/A	N/A
2016/0087486	12/2015	Pogorelik et al.	N/A	N/A
2016/0094091	12/2015	Shin et al.	N/A	N/A
2016/0094092	12/2015	Davlantès et al.	N/A	N/A
2016/0099601	12/2015	Leabman et al.	N/A	N/A
2016/0099614	12/2015	Leabman et al.	N/A	N/A
2016/0099755	12/2015	Leabman et al.	N/A	N/A
2016/0099757	12/2015	Leabman et al.	N/A	N/A
2016/0112787	12/2015	Rich	N/A	N/A
2016/0126749	12/2015	Shichino et al.	N/A	N/A

2016/0126752	12/2015	Vuori et al.	N/A	N/A
2016/0126775	12/2015	Park et al.	N/A	N/A
2016/0126776	12/2015	Kim et al.	N/A	N/A
2016/0141908	12/2015	Jakl et al.	N/A	N/A
2016/0164563	12/2015	Khawand et al.	N/A	N/A
2016/0172890	12/2015	Jeong	N/A	N/A
2016/0174162	12/2015	Nadakuduti et al.	N/A	N/A
2016/0181849	12/2015	Govindaraj	N/A	N/A
2016/0181867	12/2015	Daniel et al.	N/A	N/A
2016/0181873	12/2015	Mitcheson et al.	N/A	N/A
2016/0197522	12/2015	Zeine et al.	N/A	N/A
2016/0202343	12/2015	Okutsu	N/A	N/A
2016/0204642	12/2015	Oh	N/A	N/A
2016/0204643	12/2015	Manova-Elssibony	N/A	N/A
2016/0218545	12/2015	Schroeder et al.	N/A	N/A
2016/0233582	12/2015	Piskun	N/A	N/A
2016/0238365	12/2015	Wixey et al.	N/A	N/A
2016/0240908	12/2015	Strong	N/A	N/A
2016/0248276	12/2015	Hong et al.	N/A	N/A
2016/0294225	12/2015	Blum et al.	N/A	N/A
2016/0299210	12/2015	Zeine	N/A	N/A
2016/0301240	12/2015	Zeine	N/A	N/A
2016/0322868	12/2015	Akuzawa et al.	N/A	N/A
2016/0323000	12/2015	Liu et al.	N/A	N/A
2016/0336804	12/2015	Son et al.	N/A	N/A
2016/0339258	12/2015	Perryman et al.	N/A	N/A
2016/0344098	12/2015	Ming	N/A	N/A
2016/0359367	12/2015	Rothschild	N/A	N/A
2016/0380464	12/2015	Chin et al.	N/A	N/A
2016/0380466	12/2015	Yang et al.	N/A	N/A
2017/0005481	12/2016	Von Novak, III	N/A	N/A
2017/0005516	12/2016	Leabman et al.	N/A	N/A
2017/0005524	12/2016	Akuzawa et al.	N/A	N/A
2017/0005530	12/2016	Zeine et al.	N/A	N/A
2017/0012448	12/2016	Miller et al.	N/A	N/A
2017/0025887	12/2016	Hyun et al.	N/A	N/A
2017/0025903	12/2016	Song et al.	N/A	N/A
2017/0026087	12/2016	Tanabe	N/A	N/A
2017/0040700	12/2016	Leung	N/A	N/A
2017/0043675	12/2016	Jones et al.	N/A	N/A
2017/0047784	12/2016	Jung et al.	N/A	N/A
2017/0063168	12/2016	Uchida	N/A	N/A
2017/0077733	12/2016	Jeong et al.	N/A	N/A
2017/0077734	12/2016	Nokkonen et al.	N/A	N/A
2017/0077765	12/2016	Bell et al.	N/A	N/A
2017/0077979	12/2016	Papa et al.	N/A	N/A
2017/0085437	12/2016	Condeixa et al.	N/A	N/A
2017/0092115	12/2016	Sloo et al.	N/A	N/A
2017/0110886	12/2016	Reynolds et al.	N/A	N/A
2017/0110910	12/2016	Zeine et al.	N/A	N/A

2017/0127196	12/2016	Blum et al.	N/A	N/A
2017/0134686	12/2016	Leabman	N/A	N/A
2017/0141582	12/2016	Adolf et al.	N/A	N/A
2017/0141583	12/2016	Adolf et al.	N/A	N/A
2017/0163076	12/2016	Park et al.	N/A	N/A
2017/0168595	12/2016	Sakaguchi et al.	N/A	N/A
2017/0179763	12/2016	Leabman	N/A	N/A
2017/0187248	12/2016	Leabman	N/A	H01Q 1/02
2017/0214277	12/2016	Lee	N/A	F25D 27/005
2017/0214422	12/2016	Na et al.	N/A	N/A
2017/0274787	12/2016	Salter et al.	N/A	N/A
2017/0338695	12/2016	Port	N/A	N/A
2018/0006611	12/2017	De Jong et al.	N/A	N/A
2018/0040929	12/2017	Chappelle	N/A	N/A
2018/0048178	12/2017	Leabman	N/A	N/A
2018/0090992	12/2017	Shrivastava et al.	N/A	N/A
2018/0109146	12/2017	Meng et al.	N/A	N/A
2018/0227018	12/2017	Moshfeghi	N/A	N/A
2018/0262050	12/2017	Yankowitz	N/A	N/A
2018/0262060	12/2017	Johnston	N/A	N/A
2018/0269570	12/2017	Hosseini	N/A	N/A
2018/0269715	12/2017	Hannigan et al.	N/A	N/A
2018/0301934	12/2017	Prabhala et al.	N/A	N/A
2018/0309314	12/2017	White, II et al.	N/A	N/A
2018/0316391	12/2017	Hijikata	N/A	N/A
2018/0343040	12/2017	Luzinski et al.	N/A	N/A
2018/0375368	12/2017	Leabman et al.	N/A	N/A
2019/0052979	12/2018	Chen et al.	N/A	N/A
2019/0074862	12/2018	Wang et al.	N/A	N/A
2019/0296586	12/2018	Moshfeghi	N/A	N/A
2019/0326782	12/2018	Graham et al.	N/A	N/A
2019/0386522	12/2018	Park et al.	N/A	N/A
2019/0393729	12/2018	Contopanagos	N/A	H02J 50/40
2019/0393928	12/2018	Leabman	N/A	N/A
2020/0091608	12/2019	Alpman et al.	N/A	N/A
2020/0112204	12/2019	Hosseini et al.	N/A	N/A
2020/0235614	12/2019	Swan et al.	N/A	N/A
2020/0244111	12/2019	Johnston et al.	N/A	N/A
2020/0274397	12/2019	Hwang et al.	N/A	N/A
2020/0295599	12/2019	Heinrich et al.	N/A	N/A
2021/0034937	12/2020	Shakedd	N/A	G08B 13/2485
2021/0273684	12/2020	Yeh et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1829999	12/2005	CN	N/A
101401312	12/2008	CN	N/A

101465471	12/2008	CN	N/A
101496222	12/2008	CN	N/A
201278367	12/2008	CN	N/A
101507044	12/2008	CN	N/A
101630357	12/2009	CN	N/A
101699709	12/2009	CN	N/A
101803110	12/2009	CN	N/A
102004442	12/2010	CN	N/A
102027686	12/2010	CN	N/A
102027690	12/2010	CN	N/A
102089952	12/2010	CN	N/A
102227884	12/2010	CN	N/A
102292896	12/2010	CN	N/A
102388406	12/2011	CN	N/A
102474109	12/2011	CN	N/A
202333847	12/2011	CN	N/A
102630359	12/2011	CN	N/A
102860037	12/2012	CN	N/A
103036940	12/2012	CN	N/A
103151848	12/2012	CN	N/A
103155353	12/2012	CN	N/A
103296767	12/2012	CN	N/A
103339796	12/2012	CN	N/A
103348563	12/2012	CN	N/A
103594776	12/2013	CN	N/A
103683443	12/2013	CN	N/A
104040789	12/2013	CN	N/A
203826555	12/2013	CN	N/A
104090265	12/2013	CN	N/A
104113814	12/2013	CN	N/A
104167773	12/2013	CN	N/A
104168043	12/2013	CN	N/A
104347915	12/2014	CN	N/A
104584449	12/2014	CN	N/A
104617680	12/2014	CN	N/A
104659927	12/2014	CN	N/A
104718708	12/2014	CN	N/A
105207373	12/2014	CN	N/A
105491472	12/2015	CN	N/A
105762946	12/2015	CN	N/A
105765821	12/2015	CN	N/A
105932407	12/2015	CN	N/A
106329116	12/2016	CN	N/A
107465273	12/2016	CN	N/A
20016655	12/2001	DE	N/A
102013216953	12/2014	DE	N/A
102014219679	12/2015	DE	N/A
1028482	12/1999	EP	N/A
1081506	12/2000	EP	N/A
2346136	12/2010	EP	N/A

2397973	12/2011	EP	N/A
2545635	12/2012	EP	N/A
2747195	12/2013	EP	N/A
3067983	12/2015	EP	N/A
3118970	12/2016	EP	N/A
3145052	12/2016	EP	N/A
2404497	12/2004	GB	N/A
2556620	12/2017	GB	N/A
H 0410994	12/1991	JP	N/A
H 06327172	12/1993	JP	N/A
H 0837743	12/1995	JP	N/A
H 08103039	12/1995	JP	N/A
2000323916	12/1999	JP	N/A
2002209343	12/2001	JP	N/A
2002319816	12/2001	JP	N/A
2006157586	12/2005	JP	N/A
2006178910	12/2005	JP	N/A
2007043432	12/2006	JP	N/A
2007135335	12/2006	JP	N/A
2008092704	12/2007	JP	N/A
2008167017	12/2007	JP	N/A
2008295176	12/2007	JP	N/A
2009071835	12/2008	JP	N/A
2009525715	12/2008	JP	N/A
2009201328	12/2008	JP	N/A
2009247125	12/2008	JP	N/A
2009253649	12/2008	JP	N/A
2009290764	12/2008	JP	N/A
2010104098	12/2009	JP	N/A
2011083078	12/2010	JP	N/A
2011514781	12/2010	JP	N/A
2012016171	12/2011	JP	N/A
2012023950	12/2011	JP	N/A
2012095226	12/2011	JP	N/A
2012157167	12/2011	JP	N/A
2012170194	12/2011	JP	N/A
2013500693	12/2012	JP	N/A
2013027074	12/2012	JP	N/A
2013511908	12/2012	JP	N/A
2013099249	12/2012	JP	N/A
2013162624	12/2012	JP	N/A
2013169106	12/2012	JP	N/A
2013243908	12/2012	JP	N/A
2014011938	12/2013	JP	N/A
2014501080	12/2013	JP	N/A
2014075927	12/2013	JP	N/A
2014112063	12/2013	JP	N/A
2014135538	12/2013	JP	N/A
2014176125	12/2013	JP	N/A
2014176131	12/2013	JP	N/A

2014179746	12/2013	JP	N/A
2014217044	12/2013	JP	N/A
2014223018	12/2013	JP	N/A
2015027345	12/2014	JP	N/A
2015128349	12/2014	JP	N/A
2015128370	12/2014	JP	N/A
2015139276	12/2014	JP	N/A
2017034935	12/2016	JP	N/A
2017073855	12/2016	JP	N/A
2017077093	12/2016	JP	N/A
WO2015177859	12/2016	JP	N/A
2017085682	12/2016	JP	N/A
WO2017056260	12/2016	JP	N/A
20060061776	12/2005	KR	N/A
20070044302	12/2006	KR	N/A
100755144	12/2006	KR	N/A
100819604	12/2007	KR	N/A
20110132059	12/2010	KR	N/A
20110135540	12/2010	KR	N/A
20120009843	12/2011	KR	N/A
20120108759	12/2011	KR	N/A
20130026977	12/2012	KR	N/A
20140023409	12/2013	KR	N/A
20140023410	12/2013	KR	N/A
20140025410	12/2013	KR	N/A
20140085200	12/2013	KR	N/A
20140148270	12/2013	KR	N/A
20150077678	12/2014	KR	N/A
20160018826	12/2015	KR	N/A
20160087671	12/2015	KR	N/A
20170050971	12/2016	KR	N/A
20180108317	12/2017	KR	N/A
20180114721	12/2017	KR	N/A
20190009237	12/2018	KR	N/A
2658332	12/2017	RU	N/A
WO 199508125	12/1994	WO	N/A
WO 199831070	12/1997	WO	N/A
WO 199952173	12/1998	WO	N/A
WO 2000111716	12/2000	WO	N/A
WO 2003091943	12/2002	WO	N/A
WO 2004077550	12/2003	WO	N/A
WO 2006122783	12/2005	WO	N/A
WO 2007070571	12/2006	WO	N/A
WO 2008024993	12/2007	WO	N/A
WO 2008156571	12/2007	WO	N/A
WO 2010022181	12/2009	WO	N/A
WO 2010039246	12/2009	WO	N/A
WO 2010116441	12/2009	WO	N/A
WO 2010138994	12/2009	WO	N/A
WO 2011112022	12/2010	WO	N/A

WO 2012153529	12/2011	WO	N/A
WO 2012177283	12/2011	WO	N/A
WO 2013031988	12/2012	WO	N/A
WO 2013035190	12/2012	WO	N/A
WO 2013038074	12/2012	WO	N/A
WO 2013042399	12/2012	WO	N/A
WO 2013052950	12/2012	WO	N/A
WO 2013088238	12/2012	WO	N/A
WO 2013105920	12/2012	WO	N/A
WO 2013175596	12/2012	WO	N/A
WO 2014068992	12/2013	WO	N/A
WO 2014075103	12/2013	WO	N/A
WO 2014113093	12/2013	WO	N/A
WO 2014132258	12/2013	WO	N/A
WO 2014134996	12/2013	WO	N/A
WO 2014156465	12/2013	WO	N/A
WO 2014170773	12/2013	WO	N/A
WO 2014171348	12/2013	WO	N/A
WO 2014182788	12/2013	WO	N/A
WO 2014182788	12/2013	WO	N/A
WO 2014197472	12/2013	WO	N/A
WO 2014209587	12/2013	WO	N/A
WO 2015038773	12/2014	WO	N/A
WO 2015097809	12/2014	WO	N/A
WO 2015130902	12/2014	WO	N/A
WO 2015161323	12/2014	WO	N/A
WO 2016013944	12/2015	WO	N/A
WO 2016024869	12/2015	WO	N/A
WO 2016048512	12/2015	WO	N/A
WO 2016057115	12/2015	WO	N/A
WO 2016088261	12/2015	WO	N/A
WO 2016187357	12/2015	WO	N/A
WO 2017112942	12/2016	WO	N/A
WO 2018203176	12/2017	WO	N/A
WO 2018208130	12/2017	WO	N/A
WO 2019012372	12/2018	WO	N/A
WO 2019078660	12/2018	WO	N/A

OTHER PUBLICATIONS

Energous Corp., ISRWO, PCT/US2014/037072, Sep. 12, 2014, 8 pgs. cited by applicant
Energous Corp., Isrwo, PCT/US2014/037109, Apr. 8, 2016, 12 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/037170, Sep. 15, 2014, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/040697, Oct. 1, 2014, 12 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/040697, Dec. 8, 2015, 9 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/040705, Sep. 23, 2014, 8 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/040705, Dec. 8, 2015, 6 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/041546, Oct. 16, 2014, 12 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/041534, Oct. 13, 2014, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/046956, Nov. 12, 2014, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/049669, Nov. 13, 2014, 10 pgs. cited by applicant

Energous Corp., IPRP, PCT/US2014/049669, Feb. 9, 2016, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/046941, Nov. 6, 2014, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/062672, Jan. 26, 2015, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/040648, Oct. 10, 2014, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/041323, Oct. 1, 2014, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/044810 Oct. 21, 2014, 12 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/041558, Oct. 10, 2014, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/041342, Jan. 27, 2015, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/045119, Oct. 13, 2014, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/045119, Jan. 12, 2016, 9 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/045237, Oct. 13, 2014, 16 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/045102, Oct. 28, 2014, 14 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/046961, Nov. 24, 2014, 16 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/048002, Nov. 13, 2014, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/048002, Feb. 12, 2015 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/047963, Nov. 7, 2014, 13 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/049666, Nov. 10, 2014, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/049673, Nov. 18, 2014, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/054891, Dec. 18, 2014, 12 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/054897, Feb. 17, 2015, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/054953, Dec. 4, 2014, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/055195, Dec. 22, 2014, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/059317, Feb. 24, 2015, 13 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/059317, Apr. 12, 2016, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/059340, Jan. 15, 2015, 13 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/059340, Apr. 12, 2016, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/059871, Jan. 23, 2015, 12 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/059871, Apr. 12, 2016, 9 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/062682, Feb. 12, 2015, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/068282, Mar. 19, 2015, 13 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/068282, Jun. 7, 2016, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/068568, Mar. 20, 2015, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/068586, Mar. 20, 2015, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/068586, Jun. 14, 2016, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2014/062661, Jan. 27, 2015, 12 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2014/062661, May 3, 2016, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067242, Mar. 16, 2016, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067242, Jun. 27, 2017, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067243, Mar. 10, 2016, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067243, Jun. 27, 2017, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067245, Mar. 17, 2016, 8 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067245, Jun. 27, 2017, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067246, May 11, 2016, 18 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067246, Jun. 27, 2017, 9 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067249, Mar. 29, 2016, 8 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067249, Jun. 27, 2017, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067250, Mar. 30, 2016, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067250, Mar. 30, 2016, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067271, Mar. 11, 2016, 6 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067271, Jul. 4, 2017, 5 pgs. cited by applicant

Energous Corp., ISRWO, PCT/US2015/067275, Mar. 3, 2016, 8 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067275, 04JUL2017, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067279, Mar. 11, 2015, 13 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067279, Jul. 4, 2017, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067282, Jul. 5, 2016, 7 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067282, Jul. 4, 2017, 6 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067287, Feb. 2, 2016, 8 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067287, Jul. 4, 2017, 6 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067291, Mar. 4, 2016, 10 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067291, Jul. 4, 2017, 4 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067294, Mar. 29, 2016, 7 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067294, Jul. 4, 2017, 6 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067325, Mar. 10, 2016, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067325, Jul. 4, 2017, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2015/067334, Mar. 3, 2016, 6 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2015/067334, Jul. 4, 2017, 5 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/068495, Mar. 30, 2017, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2016/068495, Jun. 26, 2018, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/068498, May 17, 2017, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/068504, Mar. 30, 2017, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/068551, Mar. 17, 2017, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/068987, May 8, 2017, 10 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2016/068987, Jul. 3, 2018, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/068993, Mar. 13, 2017, 12 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2016/068993, Jul. 3, 2018, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/069313, Nov. 13, 2017, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2016/069316, Mar. 16, 2017, 15 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2017/046800, Sep. 11, 2017, 13 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2017/046800, Feb. 12, 2019, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2017/065886, Apr. 6, 2018, 13 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2017/065886, Jun. 18, 2019, 10 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/012806, Mar. 23, 2018, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/012806, Jul. 9, 2019, 6 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/025465, Jun. 22, 2018, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/025465, Oct. 1, 2019, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/064289, Apr. 25, 2019, 12 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/031768, Jul. 3, 2018, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/031768, Nov. 12, 2019, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/039334, Sep. 11, 2018, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/039334, Dec. 24, 2019, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/051082, Dec. 12, 2018, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/051082, Mar. 17, 2020, 9 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/031786, Aug. 8, 2018, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/031786, Apr. 14, 2020, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2019/021817, Apr. 6, 2019, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2019/021817, Sep. 15, 2020, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2018/058178, Mar. 13, 2019, 10 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2018/058178, May 5, 2020, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2019/015820, May 14, 2019, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2019/015820, Aug. 4, 2020, 7 pgs. cited by applicant

Energous Corp., ISRWO, PCT/US2019/039014, Oct. 4, 2019, 15 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2019/061445, Jan. 7, 2020, 19 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2020/067566, Apr. 27, 2021, 12 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2020/067566, Jul. 5, 2022, 8 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2020/064592, Mar. 24, 2021, 10 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2020/064592, May 17, 2022, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2021/027140, Jul. 21, 2021, 9 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2021/027140, Oct. 13, 2022, 5 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2021/045269, Dec. 7, 2021, 7 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2021/045269, Feb. 7, 2023, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2021/061430, Apr. 6, 2022, 10 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2021/061430, May 30, 2023, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2021/062452, Mar. 29, 2022, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2021/062452, Jun. 13, 2023, 7 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2022/054108, May 4, 2023, 11 pgs. cited by applicant
Energous Corp., ISRWO, PCT/US2020/027409, Jul. 24, 2020, 11 pgs. cited by applicant
Energous Corp., IPRP, PCT/US2020/027409, Sep. 28, 2021, 7 pgs. cited by applicant
Order Granting Reexamination Request, U.S. Appl. No. 90/013,793 Aug. 31, 2016, 23 pgs. cited by applicant
Ossia Inc. vs Energous Corp., PGR2016-00023—Institution Decision, Nov. 29, 2016, 29 pgs. cited by applicant
Ossia Inc. vs Energous Corp., PGR2016-00024—Institution Decision, Nov. 29, 2016, 50 pgs. cited by applicant
Ossia Inc. vs Energous Corp., PGR2016-00024—Judgement-Adverse, Jan. 20, 2017, 3 pgs. cited by applicant
Notice of Intent to Issue Reexam Certificate: U.S. Appl. No. 90/013,793 Feb. 2, 2017, 8 pgs. cited by applicant
Ossia Inc. vs Energous Corp., Declaration of Stephen B. Heppe in Support of Petition for Post-Grant Review of U.S. Pat. No. 9,124,125, PGR2016-00024, May 31, 2016, 122 pgs. cited by applicant
Ossia Inc. vs Energous Corp., Petition for Post-Grant Review of U.S. Pat. No. 9,124,125, May 31, 2016, 92 pgs. cited by applicant
Ossia Inc. vs Energous Corp., Patent Owner Preliminary Response, Sep. 8, 2016, 95 pgs. cited by applicant
Ossia Inc. vs Energous Corp., Petition for Post Grant Review of U.S. Pat. No. 9,124,125, May 31, 2016, 86 pgs. cited by applicant
Ossia Inc. vs Energous Corp., Declaration of Stephen B. Heppe in Support of Petition for Post-Grant Review of U.S. Pat. No. 9,124,125, PGR2016-00023, May 31, 2016, 144 pgs. cited by applicant
Extended European Search Report, EP20909157.8, Sep. 15, 2023, 9 pgs. cited by applicant
Adamiuk et al., “Compact, Dual-Polarized UWB-Antenna, Embedded in a Dielectric,” IEEE Transactions on Antenna and Propagation, IEEE Service Center, Piscataway, NJ, US vol. 56, No. 2, Feb. 1, 2010, 8 pgs. cited by applicant
Gill et al., “A System for Change Detection and Human Recognition in Voxel Space using the Microsoft Kinect Sensor,” 2011 IEEE Applied Imagery Pattern Recognition Workshop, 8 pgs. cited by applicant
Han et al., Enhanced Computer Vision with Microsoft Kinect Sensor: A Review, IEEE Transactions on Cybernetics vol. 43, No. 5., pp. 1318-1334, Oct. 3, 2013. cited by applicant
Hsieh et al., “Development of a Retrodirective Wireless Microwave Power Transmission System”, IEEE, 2003, pp. 393-396. cited by applicant

Leabman, "Adaptive Band-partitioning for Interference Cancellation in Communication System," Thesis Massachusetts Institute of Technology, Feb. 1997, pp. 1-70. cited by applicant

Li et al., "High-Efficiency Switching-Mode Charger System Design Considerations with Dynamic Power Path Management," Mar./Apr. 2012 Issue, 8 pgs. cited by applicant

Mao et al., "BeamStar: An Edge-Based Approach to Routing in Wireless Sensors Networks", IEEE Transactions on Mobile Computing, IEEE Service Center, Los Alamitos, CA, vol. 6, No. 11, Nov. 1, 2007, 13 pgs. cited by applicant

Mascarenas et al., "Experimental Studies of Using Wireless Energy Transmission for Powering Embedded Sensor Nodes," Nov. 28, 2009, Journal of Sound and Vibration, 13 pgs. cited by applicant

Mishra et al., "SIW-based Slot Array Antenna and Power Management Circuit for Wireless Energy Harvesting Applications", IEEE APSURSI, Jul. 2012, 2 pgs. cited by applicant

Nenzi et al., "U-Helix: On-Chip Short Conical Antenna", 7th European Conference on Antennas and Propagation (EUCAP), ISBN: 978-1-4673-2187-7, IEEE, Apr. 8, 2013, 5 pgs. cited by applicant

Qing et al., "UHF Near-Field Segmented Loop Antennas with Enlarged Interrogation Zone," 2012 IEEE International Workshop on Antenna Technology (iWAT), Mar. 1, 2012, pp. 132-135, XP055572059, ISBN: 978-1-4673-0035-3. cited by applicant

Singh, "Wireless Power Transfer Using Metamaterial Bonded Microstrip Antenna for Smart Grid WSN", 4th International Conference on Advances in Computing and Communications (ICACC), Aug. 27-29, 2014, 1 pg. cited by applicant

Smolders, "Broadband Microstrip Array Antennas", Institute of Electrical and Electronics Engineers, Digest of the Antennas and Propagation Society International Symposium, Seattle, WA, Jun. 19-24, 1994, 3 pgs. cited by applicant

Van Veen et al., "Beamforming: A Versatile Approach to Spatial Filtering", IEEE, ASSP Magazine, Apr. 1988, pp. 4-24. cited by applicant

Wei et al., "Design of a Wideband Horizontally Polarized Omnidirectional Printed Loop Antenna," IEEE Antennas and Wireless Propagation Letters, vol. 11, Jan. 3, 2012, 4 pgs. cited by applicant

Zeng et al., "A Compact Fractal Loop Rectenna for RF Energy Harvesting," IEEE Antennas And Wireless Propagation Letters, vol. 16, Jun. 26, 2017, 4 pgs. cited by applicant

Zhai et al., "A Practical Wireless Charging System Based on Ultra-Wideband Retro-Reflective Beamforming", 2010 IEEE Antennas and Propagation Society International Symposium, Toronto, ON, 2010, 4 pgs. cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/146,314, filed on Dec. 23, 2022, entitled "Small Form-Factor Devices With Integrated And Modular Harvesting Receivers, And Shelving-Mounted Wireless-Power Transmitters For Use Therewith," which is application claims priority to U.S. Provisional Application Ser. No. 63/294,555 filed on Dec. 29, 2021, entitled "Systems For Harvesting Radio Frequency Wireless Power Using One Or More Of Shelving-Mounted Wireless-Power Transmitters, Small Form-Factor Devices With Integrated And Modular Harvesting Receivers, And Wireless-Power Transmitters

Capable Of Providing Data And Power Signals In Multiple Orientations,” each of which is herein fully incorporated by reference in its respective entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to systems for harvesting radio-frequency wireless power and, in particular, to systems for harvesting RF wireless power using small-form factor devices with integrated and module harvesting receivers (e.g., the small form-factor devices can be digital price tags or small digital devices used to identify product information in warehousing settings) and shelving-mounted wireless-power transmitters. Some systems also include wireless-power transmitters capable of providing data and power signals in multiple orientations (these additional transmitters can replace or supplement the shelving-mounted wireless-power transmitters).

BACKGROUND

(2) Harvesting energy (e.g., ambient energy or energy transmitted with the intention that it be harvested by a receiver) to charge devices is gaining additional attention.

(3) Some harvesting systems require specific placements and orientations of the harvesting receiver relative to a transmitter to obtain sufficient power to make the system useful. Additionally, these systems can be tailor-made for specific electronic devices and/or charging environments, and thus have limited flexibility. Such devices are therefore poorly suited to changing environments (e.g., being moved around), environments with a variety of differing charging requirements, and struggle with devices that might have various different orientations relative to a transmitting device (both for transmitting and receiving power and data signals).

(4) Some harvesting systems also rely on the use of active beam-forming control techniques that require formation of multiple beams of energy, in which beams are formed to create focused energy in an operational area. To create this focused energy, many existing solutions use beam-forming, e.g., controlling phase and other waveform characteristics to produce constructive and/or destructive interference patterns to focus power beams onto a device at a certain location. Beam-forming typically requires multiple antennas, beam-forming algorithm control circuitry and/or algorithms, and multiple power amplifiers, all of which add complexity to the system, and add to overall system costs.

(5) As such, it would be desirable to provide systems and methods for wirelessly transmitting and harvesting wireless power that address the above-mentioned drawbacks.

SUMMARY

(6) The wireless-power transmission and harvesting system described herein solves one or more of the problems identified above by making use of one or more of three primary embodiments: (i) an RF harvesting receiver that is configured to receive radiated wireless power in multiple different orientations and which includes modular components that can be easily replaced and which are integrated with a small form-factor device (such as a digital price tag); (ii) shelving-mounted wireless-power transmitters, and (iii) wireless-power transmitters capable of providing data and power signals in multiple orientations. As one example pertaining to the second embodiment, the shelving-mounted transmitter can include a radiating antenna element that can be designed with varying numbers of conductive segments. By adjusting the number of conductive segments (which can be done at manufacture time or which can be done on-the-fly using switches or like structures to activate or disable certain conductive segments), the coverage area of a particular shelving-mounted transmitter can be adjusted. This is convenient for applications where the coverage area is not standard across all areas of an environment (e.g., an environment including multiple shelving units in a grocery store or an environment including storage shelves in a warehouse, and the like) and can fluctuate in different areas of the environment such that different required coverage areas are needed (e.g., a coverage area for multiple different shelving units in a store, or various storage shelves used in large warehouse settings). As one example with respect to the third embodiment,

the RF transmitter can be configured to radiate RF signals with a circular polarization while concurrently radiating communications signals in a dual linear fashion (e.g., one data antenna can radiate data signals with a horizontal polarization and another perpendicularly-oriented data antenna can radiate data signals with a vertical polarization). Additionally, integrating multi-orientation data and power signals in a transmitter enables further control of the receiving devices. In one example, where the receiving device is an inventory tag, the electronic device can be updated easily to reflect changes in inventory or price. Examples of these improvements are discussed in detail below, which provide solutions to one or more of the problems discussed earlier. (7) Additionally, any of the embodiments described herein can utilize transmission techniques that do not require any active beam-forming control (e.g., a single antenna can be utilized with a single power amplifier to allow for wirelessly delivering energy to a harvester device), thereby producing efficient systems with fewer components.

(8) (A1) In accordance with some embodiments, a wireless-power harvester is integrated in a small form-factor device. The wireless-power harvester comprises a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns. The first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and the second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB is configured to operate as a ground plane for the stamped metal antenna. An intermediate portion is disposed between the first end and the second end of the meandering shape coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves, harvested by the stamped metal harvesting antenna, into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

(9) (A2) In some embodiments of A1, the small form-factor device is a digital price tag that includes a display powered by the battery. The display and the battery are coupled to the PCB.

(10) (A3) In some embodiments of any of A1-A2, the small form-factor device is a digital thermometer powered by the battery.

(11) (A4) In some embodiments of any of A1-A3, the power-conversion circuitry is on a substrate distinct from the PCB.

(12) (A5) In some embodiments of any of A1-A4, the stamped metal harvesting antenna is quasi omnidirectional.

(13) (A6) In some embodiments of any of A1-A5, the stamped metal harvesting antenna is coupled to an edge of the PCB.

(14) (A7) In some embodiments of any of A1-A6, the stamped metal harvesting antenna is modular, such that it can be interchangeably coupled with the PCB and the power-conversion circuitry at a first point in time, and separately coupled with a different PCB of a different small form-factor device and different power-conversion circuitry at a second point in time that is distinct from the first point in time.

(15) (A8) In some embodiments of any of A1-A7, the stamped metal antenna occupies a first area of the housing of the small form-factor device that is adjacent to a second area of the housing occupied by the PCB, and the first and second areas are non-overlapping.

(16) (A9) In some embodiments of any of A1-A8, the RF power waves have a centering operating frequency of 918 MHz.

(17) (A10) In some embodiments of any of A1-A9, the stamped metal harvesting antenna has a gain of at least 2 dB.

(18) (A11) In some embodiments of any of A1-A10, the predetermined number of turns is two.

(19) (A12) In some embodiments of any of A1-A11, the battery has a capacity of 60 to 100 mAh.

(20) (A13) In some embodiments of any of A1-A12, the stamped metal harvesting antenna has a

thickness of 60 mils (about 1.5 mm).

(21) (A14) In some embodiments of any of A1-A13, the stamped metal harvesting antenna has a width of around 1 inch.

(22) (A15) In some embodiments of any of A1-A14, the small form-factor device includes a communications component that is coupled to the PCB, the communications component configured to receive data that allows the small form-factor device to display graphical information.

(23) (A16) In some embodiments of any of A1-A15, the graphical information is displayed using a text-only display of the small form-factor device.

(24) (B1) In another aspect, a small form-factor device, comprises a wireless-power harvester. The wireless-power harvester includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns. A first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB is configured to operate as a ground plane for the stamped metal antenna. An intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device. The wireless-power harvester of the small form-factor device of B1 can be configured in accordance with any of A2-A16. A method of operating the wireless-power harvester of any of claims A1-A16 and or the small form-factor device that includes the harvester is also contemplated, the method operates the harvester to allow it to harvest radio-frequency wireless-power waves, which are then converted to usable energy for power or charging the small form-factor device.

(25) (C1) In another aspect, a shelving-mounted wireless-power-transmitting and receiving system comprises a shelving-mounted wireless-power-transmitting device. The shelving-mounted wireless-power-transmitting device is configured to transmit RF power waves to a wireless-power harvester device that is integrated in a small form-factor device. The wireless-power harvester device integrated in the small form-factor device includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns. A first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB configured to operate as a ground plane for the stamped metal antenna. An intermediate portion, disposed between the first end and the second end, of the meandering shape is power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more of the RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

(26) (C2) In some embodiments of C1, the shelving-mounted wireless-power-transmitting device comprises a mounting structure that is mountable to a shelving unit, the shelving unit having a predetermined height and a predetermined width. The mounting includes a first end coupled to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. The second end includes: a ground plane, a first plurality of conductive segments having a first shape and a first size, each of the first plurality of conductive segments disposed over the ground plane. The second end includes a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size. A respective conductive segment of the second plurality of conductive segments separates adjacent conductive

segments of the first plurality of conductive segments from one another. The second end includes the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area of the RF wireless power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves. The harvester of C1 or C2 can be configured in accordance with any of A2-A16, and the wireless-power transmitting device can be configured in accordance with any of D2-D11.

(27) (D1) In another aspect, a shelving-mounted wireless-power-transmitting device comprises a mounting structure that is mountable to a shelving unit, the shelving unit having a predetermined height and a predetermined width. The mounting includes a first end coupled to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. The second end includes a ground plane, and a first plurality of conductive segments having a first shape and a first size, a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size. The second end includes a respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another. The second end includes the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area of the RF wireless power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves.

(28) (D2) In some embodiments of D1, each conductive segment of the first plurality of conductive segments and the second plurality of conductive segments are separated by a predetermined gap.

(29) (D3) In some embodiments of any of D1-D2, the first shape and the second shape is a meandering path that produces a rectangular shape.

(30) (D4) In some embodiments of any of D1-D3, the first size is larger than the second size.

(31) (D5) In some embodiments of any of D1-D4, the first plurality conductive segments and the second plurality of conductive segments each include a predetermined number of two or more conductive segments.

(32) (D6) In some embodiments of any of D1-D5, the outer portion of the shelving unit is a centrally located between two posts of the shelving unit.

(33) (D7) In some embodiments of any of D1-D6, respective numbers of segments in the first and second pluralities of conductive segments are adjustable to allow for creating an altered coverage area for the shelving-mounted wireless-power transmitting device.

(34) (D8) In some embodiments of any of D1-D7, a peak gain of the antenna is at least 5.5 dBi and the wireless power waves are radiated using a center operating frequency of 918 MHz.

(35) (D9) In some embodiments of any of D1-D8, the first plurality of conductive segments and the second plurality of conductive segments are formed using respective stamped copper plates.

(36) (D10) In some embodiments of any of D1-D9, the first plurality of conductive segments includes a first number of conductive segments, the first number of conductive segments selected based on the predetermined height and width of the shelving unit. The second plurality of conductive segments includes a second number of conductive segments, the second number also selected based on the predetermined height and width of the shelving unit.

(37) (D11) In some embodiments of any of D1-D10, the shelving-mounted wireless-power-transmitting device further comprises a second mounting structure that is mountable to a second shelving unit, the second shelving unit having a larger width and height than the predetermined height and width of the shelving unit, and the second mounting structure having a ground plane.

The shelving-mounted wireless-power-transmitting device further comprises a third plurality of conductive segments with each adjacent conductive segment of the third plurality is separated by a conductive segment of a fourth plurality of conductive segments. The third and fourth pluralities of conductive segments are disposed over the ground plane to form a second antenna that is configured to radiate second RF wireless power waves towards the second shelving unit such that the larger width and height of the second shelving unit is with a second coverage area of the second RF wireless power waves. A third number of conductive segments in the third plurality of conductive segments is larger than the first number of conductive segments and a fourth number of conductive segments in the fourth plurality of conductive segments is larger than the second number of conductive segments, and the second coverage area is larger than the coverage area.

(38) (E1) In another aspect, a wireless-power-transmitting device, includes a backplane that includes a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band. The RF wireless-power antenna has a perimeter that is within a perimeter of the backplane. In some embodiments, the backplane includes a first data-communications antenna configured to transmit data signals using a second frequency band distinct from the first frequency band, and the first data-communications antenna being coupled to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna. The backplane includes a second data-communications antenna configured to transmit data signals using the second frequency band, and the second data-communications antenna is coupled to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna.

(39) (E2) In some embodiments of E1, the wireless-power-transmitting device includes a spacer that is coupled between the RF wireless-power antenna and the backplane such that the RF wireless-power antenna is coupled to the spacer and sits above the backplane.

(40) (E3) In some embodiments of any of E1-E2, the RF wireless-power antenna is circularly polarized such that the wireless power waves using the first frequency band are transmitted having a circular polarization.

(41) (E4) In some embodiments of any of E1-E3, the first data communication antenna has a first polarization and the second communication antenna has a second polarization that is different from the first polarization.

(42) (E5) In some embodiments of any of E1-E4, the first and second edges of the RF wireless-power antenna are separated by a third notched edge that separates the first and second edges and is shorter in length than the first and second edges, such that the RF wireless-power antenna has a generally quadrilateral shape with at least one notched edge removed from a corner of the quadrilateral.

(43) (E6) In some embodiments of any of E1-E5, the RF wireless-power antenna has two notched edges, the two notched edges being symmetrically shaped.

(44) (E7) In some embodiments of any of E1-E6, the first communication antenna is placed in a middle of the first edge of the perimeter of the RF wireless-power antenna, and the second communication antenna is placed in a middle of the second edge of the perimeter of the RF wireless-power antenna.

(45) (E8) In some embodiments of any of E1-E7, the RF wireless-power antenna is a patch antenna.

(46) (E9) In some embodiments of any of E1-E8, the patch antenna is constructed of copper material.

(47) (E10) In some embodiments of any of E1-E9, a peak gain of the RF wireless-power antenna is greater than 8 dBi while the first frequency band is a center frequency band of 918 MHz.

(48) (E11) In some embodiments of any of E1-E10, the wireless-power antenna matching is less than -10 dB.

(49) (E12) In some embodiments of any of E1-E11, the first data-communications antenna and the second data-communications antenna produce a peak gain of 5 dBi while operating at the second frequency band of approximately 2.45 GHz.

(50) (E13) In some embodiments of any of E1-E12, the first data-communications antenna and the second data-communications antenna have a matching of greater than -7 dB.

(51) (E14) In some embodiments of any of E1-E13, the RF frequency wireless power antenna improves the gain of both the first data-communications antenna and the second data-communications antenna as a result of their proximity to the wireless-power antenna.

(52) (E15) In some embodiments of any of E1-E14, the first data-communications antenna and the second data-communications antenna use the RF frequency wireless power antenna and a main ground to provide maximum gain and optimum radiation performance of the first data-communications antenna and the second data-communications antenna.

(53) (E16) In some embodiments of any of E1-E15, the first data-communications antenna and the second data-communications antenna are formed using respective stamped copper plates.

(54) (E17) In some embodiments of any of E1-E16, the first data-communications antenna and the second data-communications antenna are each suspended 0.1 to 0.5 inches from a top surface of the backplane.

(55) (F1) In another aspect, a wireless-power transmitting and receiving system, includes a plurality of wireless power-transmitting devices including any of the shelving-mounted wireless power transmitting devices of D1-D11 or the wireless-power transmitting device of any of E1-E16, or both, a plurality of wireless-power harvesters structured in accordance with any of A1-A16, B1, and C1-C2.

(56) (G1) In another aspect, a method of manufacturing a wireless-power device comprises providing a backplane is performed. The method comprises disposing, on the backplane, a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane. The method also comprises coupling a first data-communications antenna to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna. The first data-communications antenna transmits data signals using a second frequency band distinct from the first frequency band. The method further comprises coupling a second data-communications antenna to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna. The second data-communications antenna transmits data signals using the second frequency band.

(57) Note that the various embodiments described above can be combined with any other embodiments described herein. The features and advantages described in the specification are not all inclusive and, in particular, many additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) So that the present disclosure can be understood in greater detail, a more particular description may be had by reference to the features of various embodiments, some of which are illustrated in the appended drawings. The appended drawings, however, merely illustrate pertinent features of the present disclosure. As one of skill will appreciate upon reading this disclosure in conjunction with the figures, the description may admit or contemplate other features.

- (2) FIG. 1 illustrates an example environment (e.g., a warehouse) that includes multiple of the wireless-power-transmitting devices described here and multiple wireless-power receivers that have different orientations but still harvest sufficient power, in accordance with some embodiments.
- (3) FIGS. 1A and 1B illustrate different views of a wireless-power harvester integrated in an electronic device, in accordance with some embodiments.
- (4) FIG. 2 illustrates an integrated wireless-power harvester enclosed by a housing of a small form-factor electronic device, in accordance with some embodiments,
- (5) FIGS. 3A and 3B illustrate different bottom views of an integrated wireless-power harvester, in accordance with some embodiments.
- (6) FIG. 4 illustrates a radiation pattern produced by an embodiment of the wireless-power harvester shown in FIGS. 1A and 1B, in accordance with some embodiments.
- (7) FIG. 5 illustrates a shelving-mounted wireless power transmitting system for powering and/or charging one or more wireless-power harvesters integrated into small form-factor electronic devices, in accordance with some embodiments.
- (8) FIG. 6 illustrates an example cross-section of a gain distribution (emitted radio frequency waves) from a transmitter system placed at the top of a shelving system, in accordance with some embodiments.
- (9) FIG. 7 illustrates three example variations of the shelving-mounted wireless power transmitting system for powering and/or charging one or more wireless-power harvesters integrated into small form-factor electronic devices shown in FIGS. 5-6, in accordance with some embodiments.
- (10) FIG. 8 illustrates a wireless transmitting device 800 that is configured to emit RF wireless power waves using a first frequency band and data-communication waves using a second frequency band, distinct from the first frequency band, in accordance with some embodiments.
- (11) FIG. 9 is a flow diagram showing a method of manufacturing a wireless-power device, in accordance with some embodiments.
- (12) FIG. 10 is a block diagram of components of a wireless power transmission environment, in accordance with some embodiments.
- (13) FIG. 11A is a block diagram of a wireless-power transmitter, in accordance with some embodiments.
- (14) FIG. 11B is a block diagram of another wireless-power transmitter including an RF power transmitter integrated circuit, one or more sensors, one or more antennas, and/or a power amplifier, in accordance with some embodiments.
- (15) FIG. 12 is a block diagram illustrating one or more components of a wireless-power transmitter, in accordance with some embodiments.
- (16) FIG. 13 is a block diagram illustrating a representative wireless-power receiver, in accordance with some embodiments.
- (17) In accordance with common practice, the various features illustrated in the drawings may not be drawn to scale. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may not depict all of the components of a given system, method or device. Finally, like reference numerals may be used to denote like features throughout the specification and figures.

DETAILED DESCRIPTION

(18) Numerous details are described herein in order to provide a thorough understanding of the example embodiments illustrated in the accompanying drawings. However, some embodiments may be practiced without many of the specific details, and the scope of the claims is only limited by those features and aspects specifically recited in the claims. Furthermore, well-known processes, components, and materials have not been described in exhaustive detail so as not to unnecessarily obscure pertinent aspects of the embodiments described herein.

(19) The transmitter device (also referred to as a wireless-power transmitter device or wireless-

power transmitter below) can be an electronic device that includes, or is otherwise associated with, various components and circuits responsible for, e.g., generating and transmitting electromagnetic energy, forming transmission energy within a radiation profile at locations in a transmission field, monitoring the conditions of the transmission field, and adjusting the radiation profile where needed. The radiation profile described herein refers to a distribution of energy field within the transmission range of a transmitter device or an individual antenna (also referred to as a “transmitter”). A receiver (also referred to as a wireless-power receiver) can be an electronic device that comprises at least one antenna, at least one rectifying circuit, and at least one power converter, which may utilize energy transmitted in the transmission field from a transmitter for powering or charging the electronic device.

(20) FIG. 1 illustrates an example set-up of a combination of (i) a plurality of wireless-power harvester devices that are configured to receive radiated wireless power in multiple different orientations and include modular components that can be easily replaced and which are integrated with a small form-factor device (such as a digital price tag); (ii) a plurality of shelving-mounted wireless-power transmitters, and (iii) a wireless-power transmitters capable of providing data and power signals in multiple orientations.

(21) As stated above, FIG. 1 illustrates two types of wireless-power transmitters configured to radiate RF signals to wireless-power harvester devices. Wireless-power transmitter **10** illustrates a first type of wireless-power transmitter that can be placed on walls, ceilings, or any other supporting structure within a building. The wireless-power transmitter is configured to emit RF wireless-power waves **12** to wireless-power harvester devices (e.g., wireless-power harvester device **14**) and is also configured to emit data-communication signals **16A** and **16B** (unilaterally and bidirectionally) to one or more wireless-power harvester devices (e.g., wireless-power harvester device **14**) using communication antennas **18A** and **18B** (e.g., BLE antennas). Wireless-power harvester device **14**, while shown in a certain orientation, can be placed in any other orientation and still receive sufficient power (e.g., enough power to enable the harvester to provide usable power to power or charge an associated small form-factor electronic device, which can be a digital price tag in some embodiments), due in part to the circular polarization of the RF wireless-power waves. Additionally, the wireless-power harvester device **14** can also receive data-communication signals having any orientation, due to the dual linear polarization of the data-communication signals that allows those data-communication signals to be transmitted with both horizontal and vertical polarizations. Further details regarding the first type of wireless-power transmitter are discussed below in reference to FIG. 8. Further details regarding the wireless-power harvester device are discussed in reference to FIGS. 1A-4.

(22) Shelving-mounted wireless-power transmitters **20** and **22** are a second type of wireless-power transmitter. These shelving-mounted wireless-power transmitters can be manufactured or dynamically adjusted to have different numbers of conductive segments in their respective antennas to adjust emission profiles for RF wireless-power waves. FIG. 1 shows shelving-mounted wireless-power transmitter **20** having a first number of conductive segments, which results in emission of RF power waves **24** to wireless-power harvester devices (e.g., wireless-power harvester device **28**). Shelving-mounted wireless-power transmitter **22** has a second number of conductive segments, which results in emission of RF power waves **26** to wireless-power harvester devices (e.g., wireless-power harvester device **30**). Wireless-power harvester devices **28** and **30**, while shown in a certain orientations, can be placed in any other orientation and still receive sufficient power. Additionally, the wireless-power harvester devices **28** and **30** can also receive data-communication signals in any orientation. In some embodiments, it is also possible for the wireless-power harvester devices to harvest additional RF power waves from other nearby wireless-power transmitters (e.g., wireless-power transmitter **10**). Further detail of the second type of wireless-power transmitter is discussed in reference to FIG. 5-7. Further detail regarding the wireless-power harvester device is discussed in reference to FIGS. 1A-4.

(23) FIGS. 1A and 1B illustrate different views of a wireless-power receiver (also referred to as a RF harvesting receiver) integrated in a small form-factor electronic device, in accordance with some embodiments. FIG. 1A shows a front view **100** of an electronic device **102** including a wireless-power receiver (e.g., wireless-power receiver **1020**; FIG. 10). In some embodiments, the wireless-power receiver is integrated in the electronic device **102**. The wireless-power receiver includes a harvester antenna **104** (e.g., a stamped metal harvester antenna, and/or an antennas **1024** described in reference to FIG. 10 below) coupled to a printed circuit board (PCB **108**) of the electronic device **102**. The electronic device **102** includes a display **106** powered by a power supply (e.g., a battery **1030**; FIG. 10) of the electronic device **102**. For example, FIG. 1 illustrates display **106** of each respective electronic device showing different information (e.g., temperature values for a shelf, quantity values of items for a shelf, and prices of items on a shelf, etc.). The wireless-power receiver is configured to receive wireless power transmitted by a wireless-power transmitter (e.g., transmitter **1002** of FIG. 10) such that electromagnetic energy is wirelessly transferred from the wireless-power transmitter to the wireless-power receiver, as described below. FIG. 1B shows a bottom view **150** of an electronic device **102** including a wireless-power receiver (e.g., wireless-power receiver **1020**; FIG. 10). The bottom view **150** of the electronic device **102** shows the harvester antenna **104** coupled to the PCB **108** of the electronic device **102** and a receiver integrated circuit **156**.

(24) In some embodiments, the harvester antenna **104** is a stamped metal antenna that has a meandering shape. The meandering shape includes a predetermined number of turns or curves. In some embodiments, the predetermined number of turns is at least two. The harvester antenna **104** is quasi omnidirectional and configured to receive wireless power with any polarization. In some embodiments, a position of the harvester antenna **104** within the electronic device **102** is based, in part, on a size and/or a shape of a housing enclosing the wireless-power receiver (e.g., housing of electronic device **102**). In some embodiments, the harvester antenna **104** has a thickness of 60 mils~1.5 mm (+/-0.5 mm) and a width of 25 mm (+/-15 mm), which is indicated by reference numerals **103** and **105**, respectively, in FIG. 3A. In some embodiments, the length of the harvester antenna **104** is based, in part, on the size of the electronic device **102**, the position of the harvester antenna **104** within the electronic device **102**, the number of turns included in the harvester antenna **104**, and the shape of the electronic device **102**.

(25) In some embodiments, a first end **160** of the harvester antenna **104** (e.g., the meandering shaped antenna) is a free end configured to receive one or more radio frequency (RF) power waves. In some embodiments, the harvester antenna **104** is configured to receive the RF power waves at a frequency of 918 MHz. In some embodiments, the receiving antenna **104** has a gain of at least 2 dB (shown and described below in reference to FIG. 4). In some embodiments, the harvester antenna **104** is configured to harvest RF energy transmitted in the environment from background sources other than a dedicated transmitter.

(26) In some embodiments, an intermediate portion **164** of the receiving antenna **104** is coupled to a receiver integrated circuit **156** (e.g., receiver integrated circuit **156**). As discussed below, the receiver integrated circuit **156** is configured to convert one or more RF power waves received by the first end of the meandering shape into usable energy for charging a battery of the electronic device **102** or for powering the electronic device **102**. In some embodiments, the receiver integrated circuit **156** is an integral part of the PCB **108**. Alternatively, in some embodiments, the receiver integrated circuit **156** is on a substrate distinct from the PCB **108**.

(27) In some embodiments, a second end **162** of the harvester antenna **104** is coupled to the PCB **108** of the electronic device **102**—more specifically, an edge **154** of the PCB **108**. In this way, the PCB **108** is configured to operate as a reference ground plane of the harvester antenna **104**. The second end **162** of the harvester antenna **104** can be coupled to any edge portion of the PCB **108**. In some embodiments, the edge portion of the PCB **108** at which the harvester antenna **104** is coupled is based, in part, on a size and/or a shape of a housing enclosing the wireless-power receiver (e.g.,

housing of electronic device **102**) such that the first end **160** of the harvester antenna **104** is able to receive wireless power.

(28) In some embodiments, the electronic device **102** is a digital price tag including a display **106**. The electronic device **102** can be any small form factor device. For example, the electronic device **102** can be a digital price tag, a digital timer, a digital thermometer, a digital scale, an aspect of a smart shopping cart, a tablet, a controller, or other components used to manage inventory and pricing in supermarket or warehouse contexts, etc. In some embodiments, the electronic device **102** includes a communications component (e.g., communications component **1036** and/or **1044**; FIG. **10**) configured to communicatively couple to one or more electronic devices and/or wireless-power transmitters. For example, the electronic device **102** can be located within a supermarket and communicatively couple with a plurality of wireless-power transmitters **502** positioned within the supermarket and/or a computer network within the supermarket. In some embodiments, the electronic device **102** transmits and/or receives data to the one or more communicatively coupled electronic devices and/or wireless-power transmitters **502**. The data can include charging information such as battery life, amount of power received, charge requests, etc. In some embodiments, the data can include electronic device **102** specific data. For example, a digital price tag can transmit data regarding the number of objects on a shelf, object price, total stock of an object, discount or sales related to the object, etc. Similarly, the digital price tag can receive data updating a price or other information of an object.

(29) In some embodiments, the electronic device **102** can be located within a supermarket and communicatively couple with a plurality of wireless-power transmitters **502** positioned within a warehouse for actively updating inventory levels. A plurality of electronic devices **102** and a plurality of wireless-power transmitters **502** may be positioned around the warehouse. Additionally, the plurality of electronic devices **102** and a plurality of wireless-power transmitters **502** in the warehouse can be configured to communicate with a network to update the displays of the electronic devices to reflect inventory levels. In some embodiments, the electronic devices **102** are in communication with the network. In some embodiments, plurality of electronic devices **102** communicate with the plurality of wireless-power transmitters **502**, and the plurality of wireless-power transmitters **502** communicate with the network.

(30) The harvester antenna **104** is interchangeable. More specifically, the harvester antenna **104** is configured such that it can be positioned and/or repositioned along any edge of the PCB **108** of an electronic device **102**. The harvester antenna **104** can be designed to accommodate different configurations of the electronic device **102**. For example, the receiving antenna **104** can be coupled to different electronic devices including different PCB layouts, distinct components, and/or distinct housings. The harvester antenna **104** can be retrofitted into existing electronic devices **102** and or incorporated into an electronic device **102** during manufacturing. The harvester antenna **104** is inexpensive and easy to manufacture, which increases the number of electronic devices **102** in which the harvester antenna **104** can be integrated. This increases the availability of wireless power to an electronic device **102**.

(31) FIG. **2** illustrates an integrated wireless-power receiver enclosed by an electronic device, in accordance with some embodiments.

(32) FIGS. **3A** and **3B** illustrate different perspective views of the integrated wireless-power receiver discussed in FIGS. **1A-1B** and **2**, in accordance with some embodiments. These figures show that the integrated wireless-power receiver fits within the housing of the small form-factor device, such that no changes to dimensions of the small form-factor device need be made to account for the integrated wireless-power receiver. FIG. **3A** illustrates an optional battery **148** (e.g., battery **1030** shown in FIG. **10**) placed within the housing of the small form-factor device.

(33) FIG. **4** illustrates a radiation pattern produced by an embodiment of a wireless-power receiver shown in FIGS. **1A** and **1B**. As shown, the radiation pattern **400** is substantially omnidirectional. More specifically, the wireless-power receiver has a 2 dB gain in the X, Y, and Z directions. In

some embodiments, the wireless-power receiver is able to achieve an efficiency of approximately 94%.

(34) FIG. 5 illustrates a shelving-mounted wireless-power transmitting system for powering and/or charging one or more wireless-power receiver integrated into an electronic device (e.g., electronic device **102** in FIGS. 1A and 1B), in accordance with some embodiments. In particular, FIG. 5 shows a front view **500** of a shelving system **501** (e.g., shelving rack) and a side view **502** of a shelving system **501**. Front view **500** of a shelving system **501** illustrates a shelving-mounted wireless-power transmitter system **504** placed at the top of the shelving system **501** (e.g., shelving-mounted wireless-power transmitter system **504** is placed above the one or more wireless-power receivers) that is configured to project radio frequency (RF) power waves to the one or more wireless-power receivers placed on the different shelves of the shelving system. In some embodiments, the transmitter system is mounted to the shelving system (e.g., shelving unit) via a mounting structure (e.g., mounting structure **503**). The front view **500** of the shelving system **501** also shows a plurality of electronic device(s) **102** placed on each shelf of the shelving system. The upper-most shelf **506** has a first plurality of electronic device(s) (e.g., **102(a)** through **102(o)**) placed on the upper-most shelf. In some embodiments, the first plurality of electronic device(s) are attached to items that are place on the upper most shelf. In some embodiments, the first plurality of electronic device(s) are placed directly (e.g., placed on an outward edge of the upper-most shelf) on the shelves at locations corresponding to items resting on the shelf. Front view **500** of the shelving system **501** also illustrates a middle shelf **508** that includes a second plurality of electronic device(s) **102** (e.g., **102(p)** through **102(t)**) placed on it, and a bottom shelf **510** that includes a third plurality of electronic device(s) **102** (e.g., **102(u)** through **102(y)**). Front view **500** of the shelving system **501** also shows an expanded electronic device **512**, which corresponds to the electronic devices discussed in reference to FIG. 1A through FIG. 4. FIG. 5 also illustrates a side view **502** of a shelving system **501** that shows the placement of the shelving-mounted wireless-power transmitter system **504** relative to the shelving system **501**.

(35) FIG. 6 illustrates an example cross-section **600** of the gain distribution (emitted radio frequency waves) from shelving-mounted wireless-power transmitter system **504** placed at the top of the shelving system **501**, in accordance with some embodiments. FIG. 6 also illustrates a table **602** for interpreting the gradation of the cross-section **600** of the gain.

(36) FIG. 7 illustrates three example variations of the shelving-mounted wireless power transmitting system for powering and/or charging one or more wireless-power receiver integrated into an electronic device shown in FIGS. 5-6, in accordance with some embodiments. FIG. 7 illustrates that the shelving-mounted wireless-power transmitter system **504** can be adjusted to account for varying sizes of shelving systems. In some embodiments, this adjustment is at least partially done by altering the number of conductive segments to apply coverage for varying shelving systems having different widths, depths, heights. In some embodiments, the number of conductive segments are increased (e.g., more conductive segments are added in series) in order to increase the coverage area (e.g., horizontal coverage area) of the shelving-mounted wireless-power transmitter system **504**.

(37) FIG. 7 shows a first example shelving-mounted wireless power transmitting system **700** that is configured to radiate radiofrequency (RF) wireless power waves towards the shelving unit of a first size **702** with a predetermined height **724** and a predetermined width **726**. In order to radiate RF wireless power waves to all the receivers on the shelving unit of a first size **702**, a first number of conductive segments **704** are included in the shelving-mounted wireless-power transmitter system **504**. As a result of the first number of conductive segments being included in the shelving-mounted wireless-power transmitter system **504**, a first RF coverage area **706** is produced.

(38) FIG. 7 also shows a second example shelving-mounted wireless power transmitting system **708** that is configured to radiate RF wireless power waves towards the shelving unit of a second size **710** (e.g., larger than the shelving unit of the first size **702**). In order to radiate RF wireless

power waves to all the receivers on the shelving unit of a second size **710**, a second number of conductive segments **712** are included in the shelving-mounted wireless-power transmitter system **504** (e.g., greater in number than the first number of conductive segments **704**). In some embodiments, two different pluralities of conductive segments are included, illustrated as a first plurality of conductive segments **712A** and a second plurality of conductive segments **712B** separating adjacent respective segments of the first plurality from one another by a predetermined gap distance **728**. In some embodiments, an additional shelving-mounted wireless-power transmitter system **505** can be mounted to the shelving unit. As a result of the second number of conductive segments being included in the shelving-mounted wireless-power transmitter system **504**, a second RF coverage area **714** is produced (e.g., covering more area than the first RF coverage area **706**).

(39) FIG. **7** shows a third example shelving-mounted wireless power transmitting system **716** that is configured to radiate RF wireless power waves towards the shelving unit of a third size **718** (e.g., smaller than the shelving unit of the first size **702** and the shelving unit of the second size **710**). In order to radiate RF wireless power waves to all the receivers on the shelving unit of a third size **718**, a third number of conductive segments **720** are included in the shelving-mounted wireless-power transmitter system **504** (e.g., lesser in number than the first number of conductive segments **704** and the second number of conductive segments **712**). As a result of the third number of conductive segments being included in the shelving-mounted wireless-power transmitter system **504**, a third RF coverage area **722** is produced (e.g., covering less area than the first RF coverage area **706** and second RF coverage area **714**).

(40) FIG. **8** illustrates a wireless transmitting device **800** that is configured to emit RF wireless power waves using a first frequency band and communication waves using a second frequency band, distinct from the first frequency band, in accordance with some embodiments. FIG. **8** illustrates a RF wireless-power antenna **802** that is configured to radiate wireless power waves using a first frequency band. In some embodiments, the RF wireless-power antenna **802** is configured to emit power waves in a circular polarized manner. In some embodiments, the RF wireless-power antenna **802** is notched on opposing corners (e.g., a first notch **804A** and a second notch **804B**) of the wireless-power antenna **802**. The opposing notches help aid in the circular polarization of the RF waves emitted by the RF wireless-power antenna **802**. In some embodiments, the RF wireless-power antenna **802** is coupled to a backplane **806**. In some embodiments, the RF wireless-power antenna **802** is coupled to a spacer element **808** that is coupled to the backplane **806**.

(41) FIG. **8** also illustrates that the wireless transmitting device **800** includes two data communication antennas (e.g., a first data communication antenna **810A** and a second data communication antenna **810B**) that are configured to emit communication waves using a second frequency band. The first data communication antenna **810A** and the second data communication antenna **810B** are placed perpendicular to one another on adjacent sides of the backplane **806**. In some embodiments, the first data communication antenna **810A** is placed in the middle of the first edge **812A** and the second data communication antenna **810B** is placed in the middle of the second edge **812B**. In some embodiments, the first data communication antenna **810A** and the second data communication antenna **810B** are placed around the periphery of the RF wireless-power antenna **802**. In some embodiments, the first data communication antenna **810A** is suspended by at least one post (e.g., post **814A**) and the second data communication antenna **810B** is suspended by a at least one post (e.g., post **814B**). In some embodiments, at least one post acts as a connection to the ground plane.

(42) FIG. **9** is a flow diagram showing a method of manufacturing a wireless-power device, in accordance with some embodiments. Operations (e.g., steps) of the method **900** may be performed by a manufacturer and/or a manufacturing system. In some embodiments, manufacturing a wireless-power device comprises providing (**902**) a backplane (e.g., a substrate with no computing

components). For example, a substrate with no computing components, as shown as backplane **806** in FIG. **8**.

(43) Manufacturing a wireless-power device comprises disposing (**904**), on the backplane, a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane (e.g., FIG. **8** illustrates that the RF wireless-power antenna **802** having a perimeter within a perimeter of the backplane **806**).

(44) Manufacturing a wireless-power device comprises coupling (**906**) a first data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna. The first data-communications antenna transmits data signals using a second frequency band distinct from the first frequency band (e.g., FIG. **8** illustrates first data communication antenna **810A** coupled to the backplane **806**).

(45) Manufacturing a wireless-power device comprises coupling (**908**) a second data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna. The second data-communications antenna transmit data signals using the second frequency band (e.g., FIG. **8** illustrates second data communication antenna **810B** coupled to the backplane **806** perpendicular to the first data communication antenna **810A**).

(46) FIG. **10** is a block diagram of components of wireless power transmission environment **1000**, in accordance with some embodiments. Wireless power transmission environment **1000** includes, for example, transmitters **1002** (e.g., transmitters **1002a**, **1002b** . . . **1002n**) (also referred to as wireless-power-transmitting device and shelving-mounted wireless-power-transmitting device) and one or more receivers **1020** (e.g., receivers **1020a**, **1020b** . . . **1020n**) (also referred to as wireless-power receiver and RF harvesting receiver). In some embodiments, each respective wireless power transmission environment **1000** includes a number of receivers **1020**, each of which is associated with a respective electronic device **1022**. In some instances, the transmitter **1002** is referred to herein as a “wireless-power-transmitting device” or a “wireless power transmitter.” Additionally, in some instances, the receiver **1020** is referred to herein as a “wireless-power-receiving device” or a “wireless-power receiver.”

(47) An example transmitter **1002** (e.g., transmitter **1002a**) includes, for example, one or more processor(s) **1004**, a memory **1006**, one or more antenna arrays **1010**, one or more communications components **1012** (also referred to herein as a “wireless communications radio,” a “communications radio” or simply a “radio”), and/or one or more transmitter sensors **1014**. In some embodiments, these components are interconnected by way of a communications bus **1007**.

(48) In some embodiments, a single processor **1004** (e.g., processor **1004** of transmitter **1002a**) executes software modules for controlling multiple transmitters **1002** (e.g., transmitters **1002b** . . . **1002n**). In some embodiments, a single transmitter **1002** (e.g., transmitter **1002a**) includes multiple processors **1004**, such as one or more transmitter processors (configured to, e.g., control transmission of signals **1016** by antenna array **1010**), one or more communications component processors (configured to, e.g., control communications transmitted by communications component **1012** and/or receive communications by way of communications component **1012**) and/or one or more sensor processors (configured to, e.g., control operation of transmitter sensor **1014** and/or receive output from transmitter sensor **1014**).

(49) The wireless-power receiver **1020** receives power transmission signals **1016** and/or data-communication signals **1018** transmitted by transmitters **1002**. In some embodiments, the receiver **1020** includes one or more antennas **1024** (e.g., an antenna array including multiple antenna elements), power converter **1026**, receiver sensor **1028**, and/or other components or circuitry (e.g., processor(s) **1040**, memory **1042**, and/or communication component(s) **1044**). In some embodiments, these components are interconnected by way of a communications bus **1046**.

References to these components of receiver **1020** cover embodiments in which one or more of these components (and combinations thereof) are included.

(50) The receiver **1020** converts energy from received signals **1016** (also referred to herein as RF power transmission signals, or simply, RF signals, RF waves, power waves, or power transmission signals) into electrical energy to power and/or charge electronic device **1022**. For example, the receiver **1020** uses the power converter **1026** to convert energy derived from power waves **1016** to alternating current (AC) electricity or direct current (DC) electricity to power and/or charge the electronic device **1022**. Non-limiting examples of the power converter **1026** include rectifiers, rectifying circuits, voltage conditioners, among suitable circuitry and devices.

(51) In some embodiments, the receiver **1020** is a standalone device that is detachably coupled to one or more electronic devices **1022**. For example, the electronic device **1022** has processor(s) **1032** for controlling one or more functions of the electronic device **1022**, and the receiver **1020** has processor(s) **1040** for controlling one or more functions of the receiver **1020**.

(52) In some embodiments, the receiver **1020** is a component of the electronic device **1022**. For example, processors **1032** control functions of the electronic device **1022** and the receiver **1020**. In addition, in some embodiments, the receiver **1020** includes one or more processors **1040**, which communicates with processors **1032** of the electronic device **1022**.

(53) In some embodiments, the electronic device **1022** includes one or more processors **1032**, memory **1034**, one or more communication components **1036**, and/or one or more batteries **1030**. In some embodiments, these components are interconnected by way of a communications bus **1038**. In some embodiments, communications between electronic device **1022** and receiver **1020** occur via communications component(s) **1036** and/or **1044**. In some embodiments, communications between the electronic device **1022** and the receiver **1020** occur via a wired connection between communications bus **1038** and communications bus **1046**. In some embodiments, the electronic device **1022** and the receiver **1020** share a single communications bus.

(54) In some embodiments, the receiver **1020** receives one or more power waves **1016** directly from the transmitter **1002** (e.g., via one or more antennas **1024**). In some embodiments, the receiver **1020** harvests power waves from one or more pockets of energy created by one or more power waves **1016** transmitted by the transmitter **1002**. In some embodiments, the transmitter **1002** is a near-field transmitter that transmits the one or more power waves **1016** within a near-field distance (e.g., less than approximately six inches away from the transmitter **1002**). In other embodiments, the transmitter **1002** is a far-field transmitter that transmits the one or more power waves **1016** within a far-field distance (e.g., more than approximately six inches away from the transmitter **1002**).

(55) After the power waves **1016** are received and/or energy is harvested from them, circuitry (e.g., integrated circuits, amplifiers, rectifiers, and/or voltage conditioner) of the receiver **1020** converts the energy of the power waves to usable power (i.e., electricity), which powers the electronic device **1022** and/or is stored to battery **1030** of the electronic device **1022**. In some embodiments, a rectifying circuit of the receiver **1020** translates the electrical energy from AC to DC for use by the electronic device **1022**. In some embodiments, a voltage conditioning circuit increases or decreases the voltage of the electrical energy as required by the electronic device **1022**. In some embodiments, an electrical relay conveys electrical energy from the receiver **1020** to the electronic device **1022**.

(56) In some embodiments, the electronic device **1022** obtains power from multiple transmitters **1002** and/or using multiple receivers **1020**. In some embodiments, the wireless power transmission environment **1000** includes a plurality of electronic devices **1022**, each having at least one respective receiver **1020** that is used to harvest power waves from the transmitters **1002** into power for charging the electronic devices **1022**.

(57) In some embodiments, the one or more transmitters **1002** adjust values of one or more characteristics (e.g., waveform characteristics, such as phase, gain, direction, amplitude,

polarization, and/or frequency) of power waves **1016**. For example, a transmitter **1002** selects a subset of one or more antenna elements of antenna array **1010** to initiate transmission of power waves **1016**, cease transmission of power waves **1016**, and/or adjust values of one or more characteristics used to transmit power waves **1016**. In some embodiments, the one or more transmitters **1002** adjust power waves **1016** such that trajectories of power waves **1016** converge at a predetermined location within a transmission field (e.g., a location or region in space), resulting in controlled constructive or destructive interference patterns. The transmitter **1002** may adjust values of one or more characteristics for transmitting the power waves **1016** to account for changes at the wireless-power receiver that may negatively impact transmission of the power waves **1016**.

(58) Note that, in some embodiments, the transmitter **1002** utilizes beamforming techniques to wirelessly transfer power to a receiver **1020**, while in other embodiments, the transmitter **1002** does not utilize beamforming techniques to wirelessly transfer power to a receiver **1020** (e.g., in circumstances in which no beamforming techniques are used, the transmitter controller IC **1060** discussed below might be designed without any circuitry to allow for use of beamforming techniques, or that circuitry may be present, but might be deactivated to eliminate any beamforming control capability).

(59) In some conventional systems, a viable RF power level can be provided to an optional beam-forming integrated circuit (IC) (not shown), which then provides phase-shifted signals to one or more power amplifiers. In such conventional systems, the optional beam-forming IC is used to ensure that power transmission signals sent using two or more antennas wireless-power receivers are transmitted with appropriate characteristics (e.g., phases) to ensure that power transmitted to the particular wireless-power receiver is maximized (e.g., the power transmission signals arrive in phase at the particular wireless-power receiver). The embodiments herein, however, do not always require the use of a beam-forming integrated circuit. In certain embodiments, such a beam-forming integrated circuit (and/or associated algorithm) can be included in the system, but is disabled (or is not included in the system at all) and is not used in conjunction with wirelessly-transmitted energy to receiver devices.

(60) In some embodiments, the transmitter controller IC **1060** provides the viable RF power level directly to the one or more power amplifiers and does not use any beam-forming capabilities (e.g., bypasses/disables a beam-forming IC and/or any associated algorithms if phase-shifting is not required, such as when only a single antenna **1024** is used to transmit power transmission signals to a wireless-power receiver). In some embodiments, only a single antenna is utilized with a single power amplifier (e.g., each of the transmitter devices **10**, **20**, **22**, FIG. **1**) can be configured to use a single antenna and a single power amplifier and none of the transmitter devices need make use of active beam-forming control to wirelessly deliver sufficient power to the harvesters.

(61) In some embodiments, the transmitter controller IC **1060** provides the viable RF power level directly to the one or more power amplifier units **1120** and does not use a beam-forming IC. In some embodiments, by not using beam-forming control, there is no active beam-forming control in the power transmission system. For example, in some embodiments, by eliminating the active beam-forming control, the relative phases of the power signals from different antennas are unaltered after transmission. In some embodiments, by eliminating the active beam-forming control, the phases of the power signals are not controlled and remain in a fixed or initial phase.

(62) In some embodiments, respective antenna arrays **1010** of the one or more transmitters **1002** may include a set of one or more antennas configured to transmit the power waves **1016** into respective transmission fields of the one or more transmitters **1002**. Integrated circuits of the respective transmitter **1002**, such as a controller circuit (e.g., a radio frequency integrated circuit (RFIC)) and/or waveform generator, may control the behavior of the antennas. For example, based on the information received from the receiver **1020** by way of the communication signal **1018**, a controller circuit (e.g., processor **1004** of the transmitter **1002**, FIG. **12**) may determine values of the waveform characteristics (e.g., amplitude, frequency, trajectory, direction, phase, polarization,

among other characteristics) of power waves **1016** that would effectively provide power to the receiver **1020**, and in turn, the electronic device **1022**. The controller circuit may also identify a subset of antennas from the antenna arrays **1010** that would be effective in transmitting the power waves **1016**. In some embodiments, a waveform generator circuit (not shown in FIG. **10**) of the respective transmitter **1002** coupled to the processor **1004** may convert energy and generate the power waves **1016** having the specific values for the waveform characteristics identified by the processor **1004**/controller circuit, and then provide the power waves to the antenna arrays **1010** for transmission.

(63) In some embodiments, the communications component **1012** transmits data-communication signals **1018** by way of a wired and/or wireless communication connection to the receiver **1020**. In some embodiments, the communications component **1012** generates data-communication signals **1018** used for triangulation of the receiver **1020** (e.g., test signals). In some embodiments, the data-communication signals **1018** are used to convey information between the transmitter **1002** and receiver **1020** for adjusting values of one or more waveform characteristics used to transmit the power waves **1016** (e.g., convey amounts of power derived from RF test signals). In some embodiments, the data-communication signals **1018** include information related to status, efficiency, user data, power consumption, billing, geo-location, and other types of information.

(64) In some embodiments, the communications component **1012** transmits data-communication signals **1018** to the receiver **1020** by way of the electronic device **1022a**. For example, the communications component **1012** may convey information to the communications component **1036** of the electronic device **1022a**, which the electronic device **1022a** may in turn convey to the receiver **1020** (e.g., via bus **1038**).

(65) In some embodiments, the communications component **1012** includes a communications component antenna for communicating with the receiver **1020** and/or other transmitters **1002** (e.g., transmitters **1002b** through **1002n**). In some embodiments, these data-communication signals **1018** are sent using a first channel (e.g., a first frequency band) that is independent and distinct from a second channel (e.g., a second frequency band distinct from the first frequency band) used for transmission of the power waves **1016**.

(66) In some embodiments, the receiver **1020** includes a receiver-side communications component **1044** configured to communicate various types of data with one or more of the transmitters **1002**, through a respective communication signal **1018** generated by the receiver-side communications component (in some embodiments, a respective communication signal **1018** is referred to as an advertising signal). The data may include location indicators for the receiver **1020** and/or electronic device **1022**, a power status of the device **1022**, status information for the receiver **1020**, status information for the electronic device **1022**, status information about the power waves **1016**, and/or status information for pockets of energy. In other words, the receiver **1020** may provide data to the transmitter **1002**, by way of the communication signal **1018**, regarding the current operation of the system **1000**, including: information identifying a present location of the receiver **1020** or the device **1022**, an amount of energy (i.e., usable power) received by the receiver **1020**, and an amount of power received and/or used by the electronic device **1022**, among other possible data points containing other types of information.

(67) In some embodiments, the data contained within data-communication signals **1018** is used by the electronic device **1022**, the receiver **1020**, and/or the transmitters **1002** for determining adjustments to values of one or more waveform characteristics used by the antenna array **1010** to transmit the power waves **1016**. Using a communication signal **1018**, the transmitter **1002** communicates data that is used, e.g., to identify receivers **1020** within a transmission field, identify electronic devices **1022**, determine safe and effective waveform characteristics for power waves, and/or hone the placement of pockets of energy. In some embodiments, the receiver **1020** uses a communication signal **1018** to communicate data for, e.g., alerting transmitters **1002** that the receiver **1020** has entered or is about to enter a transmission field (e.g., come within wireless-

power-transmission range of a transmitter **1002**), provide information about the electronic device **1022**, provide user information that corresponds to the electronic device **1022**, indicate the effectiveness of received power waves **1016**, and/or provide updated characteristics or transmission parameters that the one or more transmitters **1002** use to adjust transmission of the power waves **1016**.

(68) In some embodiments, the receiver **1020** does not include a distinct communications component **1044**. Rather, the receiver **1020** is configured to reflect RF signals transmitted by the transmitter **1002** at the one or more antennas **1024** and, importantly, modulate the reflected RF signals to convey data (or a message) to transmitter **1002**. In such embodiments, the transmitter **1002** may also lack a distinct communications component. Instead, the transmitter **1002** may receive the reflected RF signals at the one or more antenna arrays **1010**, and the transmitter **1002** may demodulate the reflected RF signals in order to interpret them.

(69) In some embodiments, transmitter sensor **1014** and/or receiver sensor **1028** detect and/or identify conditions of the electronic device **1022**, the receiver **1020**, the transmitter **1002**, and/or a transmission field. In some embodiments, data generated by the transmitter sensor **1014** and/or receiver sensor **1028** is used by the transmitter **1002** to determine appropriate adjustments to values of one or more waveform characteristics used to transmit the power waves **1016**. Data from transmitter sensor **1014** and/or receiver sensor **1028** received by the transmitter **1002** includes, e.g., raw sensor data and/or sensor data processed by a processor **1004**, such as a sensor processor. Processed sensor data includes, e.g., determinations based upon sensor data output. In some embodiments, sensor data received from sensors that are external to the receiver **1020** and the transmitters **1002** is also used (such as thermal imaging data, information from optical sensors, and others).

(70) FIG. **11A** is a block diagram of a wireless-power transmitter, in accordance with some embodiments. The block diagram of a wireless-power transmitter **1100** corresponds to an example of the components that can be included within the wireless-power transmitter **1002** described above in reference to FIG. **10**. The wireless-power transmitter **1002** can be referred to herein as a near-field (NF) power transmitter device, transmitter, power transmitter, or wireless-power transmitter device. The wireless-power transmitter **1002** includes one or more of one or more communications components **1110**, one or more power amplifier units **1120-1**, . . . **1120-n**, one or more power-transfer elements (e.g., such as antennas **1130-1** to **1130-n** (which can be instances of the transmitter antenna elements **1036**; FIGS. **1-12D**)), an RF Power Transmitter Integrated Circuit (RFIC) **1160** (e.g., analogous to controller **1040** FIG. **10**), and one or more sensors **1165**.

(71) In some embodiments, the communication component(s) **1110** (e.g., wireless communication components, such as WI-FI or BLUETOOTH radios) enable communication between the wireless-power transmitter **1100** and one or more communication networks. In some embodiments, the communication component(s) **1110** are capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, MiWi, etc.) custom or standard wired protocols (e.g., Ethernet, HomePlug, etc.), and/or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

(72) In some embodiments, the communication component(s) **1110** receives charging information from a wireless-power receiver (or from an electronic device configured to be charged by the wireless-power receiver; e.g., a wireless-power receiver **1055** described in reference to FIG. **13**). In some embodiments, the charging information is received in a packet of information that is received in conjunction with an indication that the wireless-power receiver is located within one meter of the wireless-power transmitter **1035** in FIG. **14**. In some embodiments, the charging information includes the location of the wireless-power receiver **1055** within the transmission field of the wireless-power transmitter **1035** (or the surrounding area within the communications component(s) range). For example, communication components **1110**, such as BLE communications paths

operating at 2.4 GHz, to enable the wireless-power transmitter **1035** to monitor and track the location of the wireless-power receiver **1055**. The location of the wireless-power receiver **1055** can be monitored and tracked based on the charging information received from the wireless-power receiver **1055** via the communications components **1110**.

(73) In some embodiments, the charging information indicates that a wireless-power receiver **1055** is authorized to receive wirelessly-delivered power from the wireless-power transmitter **1035**. More specifically, the wireless-power receiver can use a wireless communication protocol (such as BLE) to transmit the charging information as well as authentication information to the one or more integrated circuits (e.g., RFIC **1160**) of the wireless-power transmitter **1035**. In some embodiments, the charging information also includes general information such as charge requests from the receiver, the current battery level, charging rate (e.g., effectively transmitted power or electromagnetic energy successfully converted to usable energy), device specific information (e.g., temperature, sensor data, receiver requirements or specifications, and/or other receiver specific information), etc.

(74) In some instances, the communication component(s) **1110** are not able to communicate with wireless-power receivers for various reasons, e.g., because there is no power available for the communication component(s) **1110** to use for the transmission of data signals or because the wireless-power receiver itself does not actually include any communication component of its own. As such, in some embodiments, the wireless-power transmitters **1035** described herein are still able to uniquely identify different types of devices and, when a wireless-power receiver **1055** is detected, figure out if that the wireless-power receiver **1055** is authorized to receive wireless-power (e.g., by measuring impedances, reflected power, and/or other techniques).

(75) The one or more power amplifiers **1120** are configured to amplify an electromagnetic signal that is provided to the one or more antennas **1130**. In some embodiments, the power amplifier **1120** used in the power transmission system controls both the efficiency and gains of the output of the power amplifier. In some embodiments, the power amplifier used in the power transmission system is a class E power amplifier **1120**. In some embodiments, the power amplifier **1120** used in the power transmission system is a Gallium Nitride (GaN) power amplifier. In some embodiments, the wireless-power transmitters **1035** is configured to control operation of the one or more power amplifiers **1120** when they drive one or more antennas **1130**. In some embodiments, one or more of the power amplifiers **1120** are a variable power amplifier including at least two power levels. In some embodiments, a variable power amplifier includes one or more of a low power level, median power level, and high power level. As discussed below in further detail, in some embodiments, the wireless-power transmitters **1035** is configured to select power levels of the one or more power amplifiers. In some embodiments, the power (e.g., electromagnetic power) is controlled and modulated at the wireless-power transmitters **1035** via switch circuitry as to enable the wireless-power transmitters **1035** to send electromagnetic energy to one or more wireless receiving devices (e.g., wireless-power receivers **1055**) via the one or more antennas **1130**.

(76) In some embodiments, the output power of the single power amplifier **1120** is equal or greater than 2 W. In some embodiments, the output power of the single power amplifier **1120** is equal or less than 15 W. In some embodiments, the output power of the single power amplifier **1120** is greater than 2 W and less than 15 W. In some embodiments, the output power of the single power amplifier **1120** is equal or greater than 4 W. In some embodiments, the output power of the single power amplifier **1120** is equal or less than 8 W. In some embodiments, the output power of the single power amplifier **1120** is greater than 4 W and less than 8 W. In some embodiments, the output power of the single power amplifier **1120** is greater than 8 W and up to 50 W.

(77) In some embodiments, by using the single power amplifier **1120** with an output power range from 2 W to 15 W, the electric field within the power transmission range of the antenna **1130** controlled by the single power amplifier **1120** is at or below a SAR value of 1.6 W/kg, which is in compliance with the FCC (Federal Communications Commission) SAR requirement in the United

States. In some embodiments, by using a single power amplifier **1120** with a power range from 2 W to 15 W, the electric field within the power transmission range of the antenna **1130** controlled by the single power amplifier **1120** is at or below a SAR value of 2 W/kg, which is in compliance with the IEC (International Electrotechnical Commission) SAR requirement in the European Union. In some embodiments, by using a single power amplifier **1120** with a power range from 2 W to 15 W, the electric field within the power transmission range of the antenna **1130** controlled by the single power amplifier **1120** is at or below a SAR value of 0.8 W/kg. In some embodiments, by using a single power amplifier **1120** with a power range from 2 W to 15 W, the electric field within the power transmission range of the antenna **1130** controlled by the single power amplifier **1120** is at or below any level that is regulated by relevant rules or regulations. In some embodiments, the SAR value in a location of the radiation profile of the antenna decreases as the range of the radiation profile increases.

(78) In some embodiments, the radiation profile generated by the antenna controlled by a single power amplifier is defined based on how much usable power is available to a wireless-power receiver when it receives electromagnetic energy from the radiation profile (e.g., rectifies and converts the electromagnetic energy into a usable DC current), and the amount of usable power available to such a wireless-power receivers **1055** can be referred to as the effective transmitted power of an electromagnetic signal. In some embodiments, the effective transmitted power of the electromagnetic signal in a predefined radiation profile is at least 0.5 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is greater than 1 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is greater than 2 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is greater than 5 W. In some embodiments, the effective transmitted power of the signal in a predefined radiation profile is less or equal to 4 W.

(79) FIG. **11B** is a block diagram of another wireless-power transmitter **1150** (e.g., wireless-power receiver **1035**) including an RF power transmitter integrated circuit **1160**, one or more sensors **1165**, one or more antennas **1130**, and/or a power amplifier **1120** in accordance with some embodiments. For case of discussion and illustration, the other wireless-power transmitters **1150** can be an instance of the wireless-power transmitter devices described above in reference to FIGS. **1-13**, and includes one or more additional and/or distinct components, or omits one or more components. In some embodiments, the RFIC **1160** includes a CPU subsystem **1170**, an external device control interface, a subsection for DC to power conversion, and analog and digital control interfaces interconnected via an interconnection component, such as a bus or interconnection fabric block **1171**. In some embodiments, the CPU subsystem **1170** includes a microprocessor unit (CPU) **1173** with related Read-Only-Memory (ROM) **1172** for device program booting via a digital control interface, e.g., an I2C port, to an external FLASH containing the CPU executable code to be loaded into the CPU Subsystem Random Access Memory (RAM) **1174** (e.g., memory **1006**, FIG. **10**) or executed directly from FLASH. In some embodiments, the CPU subsystem **1370** also includes an encryption module or block **1176** to authenticate and secure communication exchanges with external devices, such as wireless-power receivers that attempt to receive wirelessly delivered power from the Wireless-power transmitters **1035**. In some embodiments, the wireless-power transmitters **1035** may also include a temperature monitoring circuit (not shown) that is in communication with the CPU subsystem **1170** to ensure that the wireless-power transmitters **1035** remains within an acceptable temperature range. For example, if a determination is made that the wireless-power transmitters **1035** has reached a threshold temperature, then operation of the wireless-power transmitters **1035** may be temporarily suspended until the wireless-power transmitters **1035** falls below the threshold temperature.

(80) In some embodiments, the RFIC **1160** also includes (or is in communication with) a power amplifier controller IC (PAIC) **1161A** that is responsible for controlling and managing operations of a power amplifier, including, but not limited to, reading measurements of impedance at various

measurement points within the power amplifier, instructing the power amplifier to amplify the electromagnetic signal, synchronizing the turn on and/or shutdown of the power amplifier, optimizing performance of the power amplifier, protecting the power amplifier, and other functions discussed herein. In some embodiments, the impedance measurement are used to allow the wireless-power transmitters **1035** (via the RFIC **1160** and/or PAIC **1161A**) to detect of one or more foreign objects, optimize operation of the one or more power amplifiers, assess one or more safety thresholds, detect changes in the impedance at the one or more power amplifiers, detect movement of the receiver within the wireless transmission field, protect the power amplifier from damage (e.g., by shutting down the power amplifier, changing a selected power level of the power amplifier, and/or changing other configurations of the wireless-power transmitters **1035**), classify a receiver (e.g., authorized receivers, unauthorized receivers, and/or receiver with an object), compensate for the power amplifier (e.g., by making hardware, software, and/or firmware adjustments), tune the wireless-power transmitters **1035**, and/or other functions.

(81) In some embodiments, the PAIC **1161A** may be on the same integrated circuit as the RFIC **1160**. Alternatively, in some embodiments, the PAIC **1161A** may be on its own integrated circuit that is separate from (but still in communication with) the RFIC **1160**. In some embodiments, the PAIC **1161A** is on the same chip with one or more of the power amplifiers **1120**. In some other embodiments, the PAIC **1161A** is on its own chip that is a separate chip from the power amplifiers **1120**. In some embodiments, the PAIC **1161A** may be on its own integrated circuit that is separate from (but still in communication with) the RFIC **1160** enables older systems to be retrofitted. In some embodiments, the PAIC **1161A** as a standalone chip communicatively coupled to the RFIC **1160** can reduce the processing load and potential damage from over-heating. Alternatively or additionally, in some embodiments, it is more efficient to design and use two different ICs (e.g., the RFIC **1160** and the PAIC **1161A**).

(82) In some embodiments, executable instructions running on the CPU (such as those shown in the memory **1206** in FIG. **12**, and described below) are used to manage operation of the wireless-power transmitters **1035** and to control external devices through a control interface, e.g., SPI control interface **1175**, and the other analog and digital interfaces included in the RFIC **1160**. In some embodiments, the CPU subsystem **1170** also manages operation of the subsection of the RFIC **1160**, which includes a local oscillator (LO) **1177** and a transmitter (TX) **1178**. In some embodiments, the LO **1177** is adjusted based on instructions from the CPU subsystem **1170** and is thereby set to different desired frequencies of operation, while the TX converts, amplifies, modulates the output as desired to generate a viable power level.

(83) In some embodiments, the RFIC **1360** and/or PAIC **1161A** provide the viable power level (e.g., via the TX **1178**) directly to the one or more power amplifiers **1120** and does not use any beam-forming capabilities (e.g., bypasses/disables a beam-forming IC and/or any associated algorithms if phase-shifting is not required, such as when only a single antenna **1130** is used to transmit power transmission signals to a wireless-power receiver **1055**). In some embodiments, by not using beam-forming control, there is no active beam-forming control in the power transmission system. For example, in some embodiments, by eliminating the active beam-forming control, the relative phases of the power signals from different antennas are unaltered after transmission. In some embodiments, by eliminating the active beam-forming control, the phases of the power signals are not controlled and remain in a fixed or initial phase. In some embodiments, the RFIC **1160** and/or PAIC **1161A** regulate the functionality of the power amplifiers **1120** including adjusting the viable power level to the power amplifiers **1120**, enabling the power amplifiers **1120**, disabling the power amplifiers **1120**, and/or other functions.

(84) Various arrangements and couplings of power amplifiers **1120** to antenna coverage areas **1190** allow the wireless-power receiver **1055** to sequentially or selectively activate different antenna coverage areas **1190** (i.e., power transfer points) in order to determine the most efficient and safest (if any) antenna coverage area **1190** to use for transmitting wireless-power to a wireless-power

receiver **1055**.

(85) In some embodiments, the one or more power amplifiers **1120** are also controlled by the CPU subsystem **1170** to allow the CPU **1173** to measure output power provided by the power amplifiers **1120** to the antenna coverage areas (i.e., plurality of power-transfer points **202**) of the wireless-power transmitter **1035**. In some embodiments, the one or more power amplifiers **1120** are controlled by the CPU subsystem **1170** via the PAIC **1161A**. In some embodiments, the power amplifiers **1120** may include various measurement points that allow for at least measuring impedance values that are used to enable the foreign object detection techniques, receiver and/or foreign object movement detection techniques, power amplifier optimization techniques, power amplifier protection techniques, receiver classification techniques, power amplifier impedance detection techniques, and/or other safety techniques described in commonly-owned U.S. patent application Ser. No. 16/932,631, which is incorporated by reference in its entirety for all purposes.

(86) FIG. **12** is a block diagram illustrating one or more components of a wireless-power transmitter **1035**, in accordance with some embodiments. In some embodiments, the wireless-power transmitter **1035** includes an RFIC **1160** (and the components included therein, such as a PAIC **1161A** and others described above in reference to FIGS. **11A-11B**), memory **1206** (which may be included as part of the RFIC **1160**, such as nonvolatile memory **1206** that is part of the CPU subsystem **1170**), one or more CPUs **1173**, and one or more communication buses **1208** for interconnecting these components (sometimes called a chipset). In some embodiments, the wireless-power transmitter **1035** includes one or more sensors **1165**. In some embodiments, the wireless-power transmitter **1035** includes one or more output devices such as one or more indicator lights, a sound card, a speaker, a small display for displaying textual information and error codes, etc. In some embodiments, the wireless-power transmitter **1035** includes a location detection device, such as a GPS other geo-location receiver, for determining the location of the wireless-power transmitter **1035**.

(87) In some embodiments, the one or more sensors **1165** include one or more capacitive sensors, inductive sensors, ultrasound sensors, photoelectric sensors, time-of-flight sensors (e.g., IR sensors, ultrasonic time-of-flight sensors, phototransistor receiver systems, etc.), thermal radiation sensors, ambient temperature sensors, humidity sensors, IR sensors or IR LED emitter, occupancy sensors (e.g., RFID sensors), ambient light sensors, motion detectors, accelerometers, heat detectors, hall sensors, proximity sensors, sound sensors, pressure detectors, light and/or image sensors, and/or gyroscopes, as well as integrated sensors in one or more antennas.

(88) In some embodiments, the wireless-power transmitter **1035** further includes an optional signature-signal receiving circuit **1240**, an optional reflected power coupler **1248**, and an optional capacitive charging coupler **1250**.

(89) The memory **1406** includes high-speed random access memory, such as DRAM, SRAM, DDR SRAM, or other random access solid state memory devices; and, optionally, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. The memory **1206**, or alternatively the non-volatile memory within memory **1206**, includes a non-transitory computer-readable storage medium. In some embodiments, the memory **1206**, or the non-transitory computer-readable storage medium of the memory **1406**, stores the following programs, modules, and data structures, or a subset or superset thereof: Operating logic **1216** including procedures for handling various basic system services and for performing hardware dependent tasks; Communication module **1228** for coupling to and/or communicating with remote devices (e.g., remote sensors, transmitters, receivers, servers, mapping memories, etc.) in conjunction with wireless communication component(s) **1110**; Sensor module **1220** for obtaining and processing sensor data (e.g., in conjunction with sensor(s) **1365**) to, for example, determine or detect the presence, velocity, and/or positioning of object in the vicinity of the wireless-power transmitter **1035** as well as classify a detected object; Power-wave generating module **1222** for

generating and transmitting power transmission signals (e.g., in conjunction with antenna coverage areas **1190** and the antennas **1130** respectively included therein), including but not limited to, forming pocket(s) of energy at given locations, and controlling and/or managing the power amplifier (e.g., by performing one or functions of the PAIC **1361A**). Optionally, the power-wave generating module **1222** may also be used to modify values of transmission characteristics (e.g., power level (i.e., amplitude), phase, frequency, etc.) used to transmit power transmission signals by individual antenna coverage areas; Impedance determining module **1223** for determining an impedance of the power amplifier based on parametric parameters obtained from one or more measurement points within the wireless-power transmitter **1035** (e.g., determining an impedance using one or more Smith charts). Impedance determining module **1223** may also be used to determine the presence of a foreign object, classify a receiver, detect changes in impedances, detect movement of a foreign object and/or receiver, determine optimal and/or operational impedances, as well as a number of other functions describe below; Database **1224**, including but not limited to: Sensor information **1226** for storing and managing data received, detected, and/or transmitted by one or more sensors (e.g., sensors **1165** and/or one or more remote sensors); Device settings **1228** for storing operational settings for the wireless-power transmitter **1035** and/or one or more remote devices including, but not limited to, lookup tables (LUT)s for SAR, e-field roll-off, producing a certain radiation profile from among various radiation profiles, Smith Charts, antenna tuning parameters, and/or values associated with parametric parameters of the wireless-power transmitter **1035** for different configurations (e.g., obtained during simulation, characterization, and/or manufacture tests of the wireless-power transmitter **1035** and/or updated during operation (e.g., learned improvements to the system)). Alternatively, raw values can be stored for future analysis; Communication protocol information **1230** for storing and managing protocol information for one or more protocols (e.g., custom or standard wireless protocols, such as ZigBee, Z-Wave, etc. and/or custom or standard wired protocols, such as Ethernet); and Optional learned signature signals **1232** for a variety of different wireless-power receivers and other objects (which are not wireless-power receivers). A secure element module **1234** for determining whether a wireless-power receiver is authorized to receive wirelessly delivered power from the wireless-power transmitter **1035**; An antenna zone selection and tuning module **1237** for coordinating a process of transmitting test power transmission signals to an antenna **1230** (e.g., antenna element **1036**) with various antenna coverage areas (i.e., power-transfer points) to determine which antenna coverage area (i.e., power-transfer point) should be used to wirelessly deliver power to various wireless-power receivers as described herein (additional examples and embodiments are provided in reference to FIGS. **9A-9B** of PCT Patent Application No. PCT/US2019/015820 (U.S. Pat. No. 10,615,647), which is incorporated by reference in its entirety for all purposes; and also provided in PCT/US2017/065886 (U.S. Pat. No. 10,256,677), which is incorporated by reference in its entirety for all purposes); An authorized receiver and object detection module **1238** used for detecting various signature signals from wireless-power receivers and from other objects, and then determining appropriate actions based on the detecting of the various signature signals (as is explained in more detail in reference to FIGS. **9A-9B** of PCT Patent Application No. PCT/US2019/015820 (U.S. Pat. No. 10,615,647), which is incorporated by reference in its entirety for all purposes; also explained in more detail in PCT/US2017/065886 (U.S. Pat. No. 10,256,677), which is incorporated by reference in its entirety for all purposes); and An optional signature-signal decoding module **1239** used to decode the detected signature signals and determine message or data content. In some embodiments, the module **1239** includes an electrical measurement module **1242** to collect electrical measurements from one or more receivers (e.g., in response to power beacon signals), a feature vector module **1244** to compute feature vectors based on the electrical measurements collected by the electrical measurement module **1239**, and/or machine learning classifier model(s) **1246** that are trained to detect and/or classify foreign objects (additional detail provided in commonly-owned U.S. Patent Publication No. 2019/0245389, which is incorporated by reference herein for all purposes).

(90) Each of the above-identified elements (e.g., modules stored in memory **1206** of the wireless-power transmitter **1035**) is optionally stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing the function(s) described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are optionally combined or otherwise rearranged in various embodiments. In some embodiments, the memory **1206**, optionally, stores a subset of the modules and data structures identified above.

(91) FIG. **13** is a block diagram illustrating a representative wireless-power receiver **1055** (also sometimes interchangeably referred to herein as a receiver, or power receiver), in accordance with some embodiments. In some embodiments, the wireless-power receiver **1055** includes one or more processing units (e.g., CPUs, ASICs, FPGAs, microprocessors, and the like) **1352**, one or more communication components **1354**, memory **1356**, antenna(s) **1360** (which can be instances receiver antenna elements **104**; FIGS. **1A-3B**), power harvesting circuitry **1359** (e.g., PCB **108**; FIGS. **1A-3B**), and one or more communication buses **1358** for interconnecting these components (sometimes called a chipset). In some embodiments, the wireless-power receiver **1055** includes one or more optional sensors **1362**, similar to the one or sensors **1028** and **1165** described above with reference to FIGS. **10** and **11**, respectively. In some embodiments, the wireless-power receiver **1055** includes an energy storage device **1361** for storing energy harvested via the power harvesting circuitry **1359**. In various embodiments, the energy storage device **1361** includes one or more batteries, one or more capacitors, one or more inductors, and the like.

(92) In some embodiments, the power harvesting circuitry **1359** includes one or more rectifying circuits and/or one or more power converters. In some embodiments, the power harvesting circuitry **1359** includes one or more components (e.g., a power converter) configured to convert energy from power waves and/or energy pockets to electrical energy (e.g., electricity). In some embodiments, the power harvesting circuitry **1359** is further configured to supply power to a coupled electronic device, such as a laptop or phone. In some embodiments, supplying power to a coupled electronic device include translating electrical energy from an AC form to a DC form (e.g., usable by the electronic device).

(93) In some embodiments, the optional signature-signal generating circuit **1310** includes one or more components as discussed with reference to FIGS. **3A-3D** of commonly-owned U.S. Patent Publication No. 2019/0245389, which is incorporated by reference in its entirety for all purposes.

(94) In some embodiments, the antenna(s) **1360** include one or more helical antennas, such as those described in detail in commonly-owned U.S. Pat. No. 10,734,717, which is incorporated by reference in its entirety for all purposes (e.g., with particular reference to FIGS. **2-4B**, and elsewhere).

(95) In some embodiments, the wireless-power receiver **1055** includes one or more output devices such as one or more indicator lights, a sound card, a speaker, a small display for displaying textual information and error codes, etc. In some embodiments, the wireless-power receiver **1055** includes a location detection device, such as a GPS (global positioning satellite) or other geo-location receiver, for determining the location of the wireless-power transmitter **1055**.

(96) In various embodiments, the one or more sensors **1362** include one or more thermal radiation sensors, ambient temperature sensors, humidity sensors, IR sensors, occupancy sensors (e.g., RFID sensors), ambient light sensors, motion detectors, accelerometers, and/or gyroscopes. It is noted that the foreign object detection techniques can operate without relying on the one or more sensor(s) **1362**.

(97) The communication component(s) **1354** enable communication between the wireless-power receiver **1055** and one or more communication networks. In some embodiments, the communication component(s) **1354** are capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, MiWi, etc.) custom or standard wired

protocols (e.g., Ethernet, HomePlug, etc.), and/or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document. It is noted that the foreign object detection techniques can operate without relying on the communication component(s) **1354**.

(98) The communication component(s) **1354** include, for example, hardware capable of data communications using any of a variety of custom or standard wireless protocols (e.g., IEEE 802.15.4, Wi-Fi, ZigBee, 6LoWPAN, Thread, Z-Wave, Bluetooth Smart, ISA100.11a, WirelessHART, MiWi, etc.) and/or any of a variety of custom or standard wired protocols (e.g., Ethernet, HomePlug, etc.), or any other suitable communication protocol, including communication protocols not yet developed as of the filing date of this document.

(99) The memory **1356** includes high-speed random access memory, such as DRAM, SRAM, DDR SRAM, or other random access solid state memory devices; and, optionally, includes non-volatile memory, such as one or more magnetic disk storage devices, one or more optical disk storage devices, one or more flash memory devices, or one or more other non-volatile solid state storage devices. The memory **1356**, or alternatively the non-volatile memory within memory **1356**, includes a non-transitory computer-readable storage medium. In some embodiments, the memory **1356**, or the non-transitory computer-readable storage medium of the memory **1356**, stores the following programs, modules, and data structures, or a subset or superset thereof: Operating logic **1366** including procedures for handling various basic system services and for performing hardware dependent tasks; Communication module **1368** for coupling to and/or communicating with remote devices (e.g., remote sensors, transmitters, receivers, servers, mapping memories, etc.) in conjunction with communication component(s) **1354**; Optional sensor module **1370** for obtaining and processing sensor data (e.g., in conjunction with sensor(s) **1362**) to, for example, determine the presence, velocity, and/or positioning of the wireless-power receiver **1055**, a wireless-power transmitter **1055**, or an object in the vicinity of the wireless-power transmitter **1055**; Wireless power-receiving module **1372** for receiving (e.g., in conjunction with antenna(s) **1360** and/or power harvesting circuitry **1359**) energy from, capacitively-conveyed electrical signals, power waves, and/or energy pockets; optionally converting (e.g., in conjunction with power harvesting circuitry **1359**) the energy (e.g., to direct current); transferring the energy to a coupled electronic device; and optionally storing the energy (e.g., in conjunction with energy storage device **1361**); Database **1374**, including but not limited to: Sensor information **1376** for storing and managing data received, detected, and/or transmitted by one or more sensors (e.g., sensors **1362** and/or one or more remote sensors); Device settings **1378** for storing operational settings for the wireless-power transmitter **1055**, a coupled electronic device, and/or one or more remote devices; and Communication protocol information **1380** for storing and managing protocol information for one or more protocols (e.g., custom or standard wireless protocols, such as ZigBee, Z-Wave, etc. and/or custom or standard wired protocols, such as Ethernet); A secure element module **1382** for providing identification information to the wireless-power transmitter **1035** (e.g., the wireless-power transmitter **1035** uses the identification information to determine if the wireless-power receiver **1304** is authorized to receive wirelessly delivered power); and An optional signature-signal generating module **1383** used to control (in conjunction with the signature-signal generating circuit **1310**) various components to cause impedance changes at the antenna(s) **1360** and/or power harvesting circuitry **1359** to then cause changes in reflected power as received by a signature-signal receiving circuit **1240**.

(100) Each of the above-identified elements (e.g., modules stored in memory **1356** of the receiver **1304**) is optionally stored in one or more of the previously mentioned memory devices, and corresponds to a set of instructions for performing the function(s) described above. The above-identified modules or programs (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules are optionally combined or otherwise rearranged in various embodiments. In some embodiments, the

memory **1356**, optionally, stores a subset of the modules and data structures identified above. Furthermore, the memory **1356**, optionally, stores additional modules and data structures not described above, such as an identifying module for identifying a device type of a connected device (e.g., a device type for an electronic device that is coupled with the receiver **1304**).

(101) In some embodiments, the near-field power transmitters disclosed herein may use adaptive loading techniques to optimize power transfer. Such techniques are described in detail in commonly-owned and incorporated-by-reference PCT Application No. PCT/US2017/065886 and, in particular, in reference to FIGS. 5-8 and 12-15 of PCT Application No. PCT/US2017/065886.

(102) In some embodiments, the wireless-power transmitter **1055** is coupled to or integrated with an electronic device, such as shelving unit, a pen, a marker, a phone, a tablet, a laptop, a hearing aid, smart glasses, headphones, computer accessories (e.g., mouse, keyboard, remote speakers), and/or other electrical devices. In some embodiments, the wireless-power transmitter **1055** is coupled to or integrated with small consumer device, such as a fitness band, a smart watch, and/or other wearable product. Alternatively, in some embodiments, the wireless-power transmitter **1055** is an electronic device.

(103) In accordance with some embodiments, a wireless-power harvester integrated in a small form-factor device (e.g., less than or equal to 100 mm in length, 70 mm in width and 30 mm in depth) (e.g., 50-100 mm in length, 35-70 mm in width, and 15-30 mm in depth) (e.g., an example small form-factor device such as the electronic device **102** shown in FIGS. **1A-1B**, **2**, and **3A-3B**), comprises a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape (e.g., FIGS. **1A-1B**, **2**, and **3A-3B** each illustrate harvester antenna **104** that follows a meandering path). The meandering shape includes a predetermined number of turns (e.g., FIGS. **1A-1B**, **2**, and **3A-3B** each illustrate harvester antenna **104** having at least two turns). Specifically, FIG. **1B** shows in a bounding box a first turn **161**. A first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB configured to operate as a ground plane for the stamped metal antenna (e.g., FIGS. **1B** and **2** illustrate harvester antenna **104** coupled to the PCB **108** of the electronic device **102** and a receiver integrated circuit **156**). An intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB, the power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device (e.g., FIGS. **1B** and **2** illustrate an intermediate portion **164** of the receiving antenna **104** being coupled to a receiver integrated circuit **156**, and FIG. **10** illustrates a battery **1030**).

(104) In some embodiments, the small form-factor device is a digital price tag including a display (e.g., an e-ink display, an LCD display, etc.) powered by the battery. The display and the battery being coupled to the PCB. For example, FIG. **1A** illustrates that the electronic device **102** includes a display **106** that is configured to be a digital price tag.

(105) In some embodiments, the small form-factor device is a digital thermometer powered by the battery. In some embodiments, the thermometer can include a display and/or provide data to a computer or server. In some embodiments, other measurement or stock tracking devices can be used. For example, a scale, a counter (e.g., identifying a number of object on a shelf). In some embodiments, shopping carts with displays can also be included.

(106) In some embodiments, the power-conversion circuitry is on a substrate distinct from the PCB.

(107) In some embodiments, the stamped metal harvesting antenna is quasi omnidirectional. For example, FIG. **1B** illustrates that the harvester antenna **104** is quasi omnidirectional and configured to receive wireless power with any polarization. In some embodiments, the antenna is configured to

receive wireless power with any polarization.

(108) In some embodiments, the stamped metal harvesting antenna is coupled to an edge of the PCB. For example, FIGS. 1B and 2 illustrate harvester antenna **104** coupled to the PCB **108** of the electronic device **102** and a receiver integrated circuit **156**. In some embodiments, the harvesting antenna can be attached to any edge portion of the PCB.

(109) In some embodiments, the stamped metal harvesting antenna is modular, such that it can be interchangeably coupled with the PCB and the power-conversion circuitry at a first point in time, and separately coupled with a different PCB of a different small form-factor device and different power-conversion circuitry at a second point in time that is distinct from the first point in time. In some embodiments, the harvesting antenna can be designed to accommodate any configuration of the small form-factor device. In some embodiments, the harvesting antenna is easy and inexpensive to manufacture. In some embodiments, the harvesting antenna can be installed by either the manufacturer of the harvesting antenna or the purchaser of the harvesting antenna (e.g., the client).

(110) In some embodiments, the stamped metal antenna occupies a first area of the housing of the small form-factor device that is adjacent to a second area of the housing occupied by the PCB, and the first and second areas are non-overlapping. For example, FIGS. 1B, 2, and 3A-3B illustrates the PCB **108** being in a non-overlapping adjacent location to the antenna **104**. Specifically, FIG. 1B shows a first area **109** that contains the PCB **108**, and a second area **111** that contains the antenna **104** and related componentry.

(111) In some embodiments, the RF power waves have a centering operating frequency of 918 MHz.

(112) In some embodiments, the stamped metal harvesting antenna has a gain of at least 2 dB. For example, FIG. 4 illustrates a harvesting antenna radiation pattern having a gain of at least 2 dB.

(113) In some embodiments, the predetermined number of turns is two. For example, FIGS. 1A-1B, 2, and 3A-3B each illustrate harvester antenna **104** having at least two turns.

(114) In some embodiments, the battery is a 60 to 100 mAh battery (e.g., battery **1030** in FIG. 10).

(115) In some embodiments, the stamped metal harvesting antenna has a thickness of 1 mm to 2 mm.

(116) In some embodiments, the stamped metal harvesting antenna has a width of 0.5 inch to 2 inches.

(117) In some embodiments, the small form-factor device includes a communications component that is coupled to the PCB, the communication component configured to receive data that allows the small form-factor device to display graphical information. For example, electronic device **102** includes a communications component (e.g., communications component **1036** and/or **1044**; FIG. 10). In some embodiments, the data regarding the number of objects on a shelf is displayed. In some embodiments, the display can display information for updating a price or other information (e.g., product name or description). In some embodiments, the display can be used to display an estimate of the price of a virtual shopping cart/list of a shopper in front of the display.

(118) In some embodiments, the graphical information is displayed using a text-only display of the small form-factor device. In some embodiments, the small form-factor device is further configured to communicatively couple to a wireless-power transmitter and provide charging information (e.g., battery life, amount of power received, charge requests, etc.).

(119) In another aspect, a small form-factor device (e.g., less than or equal to 100 mm in length, 70 mm in width and 30 mm in depth) (e.g., 50-100 mm in length, 35-70 mm in width, and 15-30 mm in depth), comprises a wireless-power harvester. The wireless-power harvester includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape. The meandering shape includes a predetermined number of turns, a first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device.

The PCB is configured to operate as a ground plane for the stamped metal antenna. In some embodiments, an intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.

(120) In another aspect, a shelving-mounted wireless-power-transmitting device is configured to transmit RF power waves to a wireless-power harvester device that is integrated in a small form-factor device. For example, FIG. 5 illustrates a front view 500 of a shelving system 501 that includes a shelving-mounted wireless-power transmitter system 504 placed at the top of the shelving system 501 for transmitting RF power waves to one or more electronic devices 102. In some embodiments, the wireless-power harvester device integrated in the small form-factor device (e.g., less than or equal to 100 mm in length, 70 mm in width and 30 mm in depth) (e.g., 50-100 mm in length, 35-70 mm in width, and 15-30 mm in depth) (e.g., an example small form-factor device such as the electronic device 102 shown in FIGS. 1A-1B, 2, and 3A-3B), includes a stamped metal harvesting antenna configured to harvest radio frequency (RF) power waves, the stamped metal antenna formed into a meandering shape (e.g., FIGS. 1A-1B, 2, and 3A-3B each illustrate harvester antenna 104 that follows a meandering path). The meandering shape of the stamped metal harvesting antenna includes a predetermined number of turns. The stamped metal harvesting antenna includes a first end of the meandering shape is a free end positioned within free space of a housing of a small form-factor device, and a second end of the meandering shape is coupled to a PCB that includes electrical components for operating and powering the small form-factor device. The PCB configured to operate as a ground plane for the stamped metal antenna (e.g., FIGS. 1B and 2 illustrate harvester antenna 104 coupled to the PCB 108 of the electronic device 102 and a receiver integrated circuit 156). An intermediate portion, disposed between the first end and the second end, of the meandering shape is coupled to power-conversion circuitry that is separate from the PCB. The power-conversion circuitry is configured to convert the one or more RF power waves harvested by the stamped metal harvesting antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device (e.g., FIGS. 1B and 2 illustrate an intermediate portion 164 of the receiving antenna 104 being coupled to a receiver integrated circuit 156, and FIG. 10 illustrates a battery 1030).

(121) In some embodiments, the shelving-mounted wireless-power-transmitting device comprises a mounting structure (e.g., mounting structure 503 shown at least in FIG. 5) that is mountable to a shelving unit (e.g., shelving system 501 shown in FIGS. 5-7). The shelving unit has a predetermined height and a predetermined width. The mounting includes a first end coupled (or integrated) to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. For example, FIGS. 5-7 illustrate shelving-mounted wireless power transmitting systems with shelving systems having predetermined height and widths. The second end includes a ground plane (dimensions of the ground plane are based on the size of the transmitting device; e.g., 8 inches by 2 inches). A first plurality of conductive segments has a first shape and a first size (e.g., FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments 704, a second number of conductive segments 708, and third number of conductive segments 720)), each of the first plurality of conductive segments is disposed over the ground plane. A second plurality of conductive segments have a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size. A respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another (e.g., a first number of conductive segments 704, a second number of conductive segments 708, and third number of conductive segments 720). The first and second pluralities of conductive segments are disposed over the

ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area (e.g., up to 15 feet in a first direction (e.g., a horizontal direction and up to 20 feet in a second direction (e.g., a vertical direction along the height of the shelving unit) of the RF wireless power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves. For example, at least enough RF energy to allow for a harvester to harvest enough usable power and/or charge for operating a digital price tag (e.g., an e-ink display and associated hardware). For example, FIG. 7 shows that the first coverage area **706**, the second coverage area **714**, and the third coverage area **722** each show the bottom shelf receiving at least a threshold amount of RF energy. In some embodiments, the usable power can be between 0.5 milliwatts and to over few tenths of Watt.

(122) In accordance with some embodiments, a shelving-mounted wireless-power-transmitting device comprises a mounting structure (e.g., mounting structure **503** shown at least in FIG. 5) that is mountable to a shelving unit (e.g., shelving system **501** shown in FIGS. 5-7). The shelving unit has a predetermined height and a predetermined width. The mounting includes a first end coupled or integrated to an outer portion of the shelving unit and a second end opposite the first end extending a predetermined distance away from the outer portion of the shelving unit. For example, FIGS. 5-7 illustrate shelving-mounted wireless power transmitting systems with shelving systems having predetermined height and widths. The second end includes a ground plane (e.g., dimensions of the ground plane are based on the size of the transmitting device; e.g., 8 inches by 2 inches). The second end includes a first plurality of conductive segments having a first shape and a first size. For example, FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**). The second end includes a second plurality of conductive segments that have a second shape and a second size, the second shape is distinct from the first shape and the second size is distinct from the first size. A respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another. For example, a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**. In some embodiments, the second end includes the first and second pluralities of conductive segments that are disposed over the ground plane to form an antenna that is configured to radiate radio-frequency (RF) wireless power waves towards the shelving unit such that (i) the predetermined height and the predetermined width of the shelving unit are within a coverage area (e.g., up to 15 feet in a first direction (e.g., a horizontal direction and up to 20 feet in a second direction (e.g., a vertical direction) of the RF wireless power waves, and (ii) a bottom shelf of the shelving unit receives at least a threshold amount of RF energy from the RF wireless power waves (e.g., at least enough RF energy to allow for a harvester to harvest enough usable power and/or charge for operating a digital price tag (e.g., an e-ink display and associated hardware)). For example, FIG. 7 shows that the first coverage area **706**, the second coverage area **714**, and the third coverage area **722** each show the bottom shelf receives at least a threshold amount of RF energy. In some embodiments, the usable power can be between 0.5 milliwatts and few tenths of a watt.

(123) In some embodiments, each conductive segment of the first plurality of conductive segments and the second plurality of conductive segments are separated by a predetermined gap. In some embodiments, the gap is between 0.1 inches and 0.5 inches. The gap distance is determined based on the predetermined height and the predetermined width of the shelving structure. In some embodiments, each segment of the first plurality of conductive segments or each segment of the second plurality of conductive segments are coupled together via a feedline. In some embodiments, each conductive segment has a length and width. In some embodiments, the conductive segment length is approximately 1.5 inches and the antenna element width is approximately 0.5 inches. In

some embodiments, the conductive segment length and width is determined based on the predetermined height and the predetermined width of the shelving unit's structure.

(124) In some embodiments, the first shape and the second shape is a meandering path that produces a rectangular shape (e.g., the first shape is a meandering 'U' shape as shown in FIG. 7 as first number of conductive segments **704**, second number of conductive segments **708**, and third number of conductive segments **716**). In some embodiments, the conductive segments are a continuous piece of metal.

(125) In some embodiments, the first size is larger than the second size. In some embodiments, the size is adjusted depending on the required coverage area (e.g., a larger shelf coverage area corresponds to a larger sized plurality of conductive segments).

(126) In some embodiments, the first plurality conductive segments and the second plurality of conductive segments each include a predetermined number of two or more conductive segments (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**). In some embodiments, the predetermined number of conductive segments is between 5 to 15 antenna elements. In some embodiments, more than 15 antenna elements can be used. In some embodiments, the predetermined number of antenna elements is based on the predetermined height and the predetermined width of the shelving unit's structure.

(127) In some embodiments, the outer portion of the shelving structure is a centrally located between two posts of the shelving structure. For example, FIG. 5 and FIG. 7 show the outer portion (e.g., mounting structure **503**) of the shelving structure being centrally located.

(128) In some embodiments, respective numbers of segments in the first and second pluralities of conductive segments are adjustable to allow for creating an altered coverage area for the shelving-mounted wireless-power transmitting device.

(129) In some embodiments, a peak gain of the antenna is at least 5.5 dBi and the wireless power waves are radiated using is a center operating frequency of 918 MHZ (e.g., FIG. 6 shows the peak gain being at least 5.5 dBi).

(130) In some embodiments, the first plurality of conductive segments and the second plurality of conductive segments are formed using respective stamped copper plates.

(131) In some embodiments, the first plurality of conductive segments includes a first number of conductive segments, the first number of conductive segments selected based on the predetermined height and width of the shelving unit. For example, FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**). The second plurality of conductive segments includes a second number of conductive segments, the second number also selected based on the predetermined height and width of the shelving unit. For example, FIG. 7 shows conductive segments of three different sizes (e.g., a first number of conductive segments **704**, a second number of conductive segments **708**, and third number of conductive segments **720**).

(132) In some embodiments, shelving-mounted wireless-power-transmitting device further comprises a second mounting structure that is mountable to a second shelving unit, the second shelving unit having a larger width and height than the predetermined height and width of the shelving unit, and the second mounting structure having a ground plane, a third plurality of conductive segments with each adjacent conductive segment of the third plurality is separated by a conductive segment of a fourth plurality of conductive segments. The third and fourth pluralities of conductive segments are disposed over the ground plane to form a second antenna that is configured to radiate second RF wireless power waves towards the second shelving unit such that the larger width and height of the second shelving unit is with a second coverage area of the second RF wireless power waves. A third number of conductive segments in the third plurality of conductive segments is larger than the first number of conductive segments and a fourth number of conductive segments in the fourth plurality of conductive segments is larger than the second

number of conductive segments, and the second coverage area is larger than the coverage area. (133) In some embodiments, to create a larger coverage area more conductive segments are added. Thereby making mounting structures that have enough conductive segments to cover the entire structure of each particular shelving unit. While the example here is of two different shelving units with different mounting structures having wireless-power transmitters with different conductive segment numbers appropriate for the dimensions of those shelving units, third, fourth, fifth sixth, etc. different numbers of such mounting structures to accommodate for different shelving structures are also contemplated.

(134) In another aspect, a backplane (e.g., a substrate with no computing components, as shown as backplane **806** in FIG. **8**) includes a radio frequency (RF) wireless-power antenna (e.g., a patch antenna, optionally with a single feed) that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane (e.g., FIG. **8** illustrates that the RF wireless-power antenna **802** having a perimeter within a perimeter of the backplane **806**). The backplane includes a first data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) configured to transmit data signals using a second frequency band distinct from the first frequency band, and the first data-communications antenna being coupled to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna (e.g., FIG. **8** illustrates first data communication antenna **810A** coupled to the backplane **806**). The backplane includes a second data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) configured to transmit data signals using the second frequency band, and the second data-communications antenna being coupled to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna. The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna (e.g., FIG. **8** illustrates second data communication antenna **810B** coupled to the backplane **806** perpendicular to the first data communication antenna **810A**).

(135) In some embodiments, the wireless-power-transmitting device includes a spacer that is coupled between the RF wireless-power antenna and the backplane such that the RF wireless-power antenna is coupled to the spacer and sits above the backplane (e.g., beneath the RF wireless-power antenna and above the backplane). For example, FIG. **8** illustrates a spacer element **808**.

(136) In some embodiments, the RF wireless-power antenna is circularly polarized such that the wireless power waves using the first frequency band are transmitted having a circular polarization (e.g., each point in a produced electromagnetic field of the wave has a constant magnitude and rotates at a constant rate in a plane perpendicular to the direction of the wave).

(137) In some embodiments, the first data communication antenna has a first polarization (e.g., horizontal polarization) and the second communication antenna has a second polarization that is different from the first polarization (e.g., a vertical polarization). In some embodiments, the first data communication antenna with the first polarization transmits data data-communication signals to receiving devices having data communication antennas with the first polarization and the second data communication antenna with the second polarization transmits data data-communication signals to receiving devices having data communication antennas with the second polarization (e.g., in this way, receiving devices can be positioned in many different orientations yet still receive reliable data data-communication signals. For instance, wireless power receiving device is oriented such that it receives horizontally polarized data signals and wireless power receiving device is oriented such that it receives vertically polarized data signals).

(138) In some embodiments, the first and second edges of the RF wireless-power antenna are separated by a third notched edge that separates the first and second edges and is shorter in length than the first and second edges, such that the RF wireless-power antenna has a generally quadrilateral shape (e.g., a shape where at least four of the sides are major sides (e.g., a first length), and additional other sides that are minor sides (e.g., having a second length less than the first length)) with at least one notched edge (e.g., a triangular cut out that alters the exterior shape).

of the quadrilateral shape) removed from a corner of the quadrilateral (e.g., a square or rectangle antenna would have the upper-left corner notched (e.g., a triangular notch) and the bottom-right corner notched). For example, FIG. 8 illustrates two notched edges (e.g., a first notch **804A** and a second notch **804B**).

(139) In some embodiments, the RF wireless-power antenna has two notched edges, the two notched edges being symmetrically shaped (e.g., FIG. 8 illustrates that the first notch **804A** and the second notch **804B** are symmetrical to each other).

(140) In some embodiments, the first communication antenna is placed in a middle of the first edge of the perimeter of the RF wireless-power antenna, and the second communication antenna is placed in a middle of the second edge of the perimeter of the RF wireless-power antenna. FIG. 8 illustrates that first data communication antenna **810A** is placed in the middle of the first edge **812A** of the perimeter of the RF wireless-power antenna, and the second data communication antenna **810B** is placed in the middle of the second edge **812B** of the perimeter of the RF wireless-power antenna.

(141) In some embodiments, wherein the RF wireless-power antenna is a patch antenna. In some embodiments, the RF wireless-power antenna is a stamped metal patch antenna.

(142) In some embodiments, the patch antenna is constructed of copper material. In some embodiments, the patch antenna is an alloy that includes copper, steel, aluminum.

(143) In some embodiments, a peak gain of the RF wireless-power antenna is greater than 8 dBi while the first frequency band is a center frequency band of 918 MHz.

(144) In some embodiments, the wireless-power antenna matching is less than -10 dB.

(145) In some embodiments, the first data-communications antenna and the second data-communications antenna produce a peak gain of 5 dBi while operating at the second frequency band of approximately 2.45 GHz. In some embodiments, the first data-communications antenna and second data-communications antenna independently produce a peak gain of 5 dBi at 2.45 GHz.

(146) In some embodiments, the first data-communications antenna and the second data-communications antenna have a matching of greater than -7 dB.

(147) In some embodiments, the RF frequency wireless power antenna improves the gain of both the first data-communications antenna and the second data-communications antenna as a result of their proximity to the wireless-power antenna (e.g., a gain of 1 dB before being near wireless-power antenna and gain of 4 dB after being near wireless-power antenna).

(148) In some embodiments, the first data-communications antenna and the second data-communications antenna use the RF frequency wireless power antenna and a main ground to provide maximum gain and optimum radiation performance of the first data-communications antenna and the second data-communications antenna.

(149) In some embodiments, the first data-communications antenna and the second data-communications antenna are formed using respective stamped copper plates.

(150) In some embodiments, the first data-communications antenna and the second data-communications antenna are each suspended 0.1 to 0.5 inches from a top surface of the backplane (e.g., FIG. 8 illustrates that the first data communication antenna **810A** is suspended by post **814A** and the second data communication antenna **810B** is suspended by post **814B**). In some embodiments, the first data-communications antenna and the second data-communications antenna are suspended by a metallic pin/post coupled to the backplane.

(151) In another aspect, a wireless-power transmitting and receiving system, includes a plurality of wireless power-transmitting devices including any of the shelving-mounted wireless power transmitting devices discussed above, and including a plurality of wireless-power harvesters structured in accordance with any of discussion above.

(152) In another aspect, a method of manufacturing a wireless-power device comprises providing a backplane is performed (e.g., a substrate with no computing components) (e.g., a substrate with no computing components, as shown as backplane **806** in FIG. 8). The method of manufacturing the

wireless-power device comprises disposing, on the backplane, a radio frequency (RF) wireless-power antenna that is configured to radiate wireless power waves using a first frequency band, the RF wireless-power antenna having a perimeter that is within a perimeter of the backplane (e.g., FIG. 8 illustrates that the RF wireless-power antenna **802** having a perimeter within a perimeter of the backplane **806**). The method of manufacturing the wireless-power device comprises coupling a first data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a first edge of the perimeter of the RF wireless-power antenna, wherein the first data-communications antenna transmits data signals using a second frequency band distinct from the first frequency band (e.g., FIG. 8 illustrates first data communication antenna **810A** coupled to the backplane **806**). The method of manufacturing the wireless-power device comprises coupling a second data-communications antenna (e.g., a Bluetooth Low Energy (BLE) radio) to the backplane such that it is outside of a second edge of the perimeter of the RF wireless-power antenna, The second edge of the perimeter of the RF wireless-power antenna is perpendicular to the first edge of the perimeter of the RF wireless-power antenna, and further wherein the second data-communications antenna transmit data signals using the second frequency band (e.g., FIG. 8 illustrates second data communication antenna **810B** coupled to the backplane **806** perpendicular to the first data communication antenna **810A**).

(153) The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the embodiments described herein and variations thereof. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the subject matter disclosed herein. Thus, the present disclosure is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

(154) Features of the present invention can be implemented in, using, or with the assistance of a computer program product, such as a storage medium (media) or computer readable storage medium (media) having instructions stored thereon/in which can be used to program a processing system to perform any of the features presented herein. The storage medium (e.g., memory **1006** and **1034** in FIG. 10) can include, but is not limited to, high-speed random access memory, such as DRAM, SRAM, DDR RAM or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, or other non-volatile solid state storage devices. Memory optionally includes one or more storage devices remotely located from the CPU(s) (e.g., processor(s)). Memory, or alternatively the non-volatile memory device(s) within the memory, comprises a non-transitory computer readable storage medium.

(155) Stored on any one of the machine readable medium (media), features of the present invention can be incorporated in software and/or firmware for controlling the hardware of a processing system (such as the components associated with the wireless-power transmitter **1035** and/or wireless-power receivers **1055**), and for enabling a processing system to interact with other mechanisms utilizing the results of the present invention. Such software or firmware may include, but is not limited to, application code, device drivers, operating systems, and execution environments/containers.

(156) It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another.

(157) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the claims. As used in the description of the embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or

more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. (158) As used herein, the term “if” may be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting,” that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” may be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.

(159) The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or to limit the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain principles of operation and practical applications, to thereby enable others skilled in the art.

Claims

1. A wireless-power transmitting system, comprising: a wireless-power transmitter coupled within an enclosed environment, wherein: the enclosed environment includes one or more shelving units, and the wireless-power transmitter is configured to radiate radio-frequency (RF) wireless-power waves towards the one or more shelving units using a first frequency band, the wireless-power transmitter includes: a backplane with one or more data-communications antennas configured to transmit data signals using a second frequency band distinct from the first frequency band, a ground plane, a first plurality of conductive segments having a first shape and a first size, a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size, wherein a respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another, and the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate RF wireless-power waves; and a wireless-power receiver integrated in a small form-factor device coupled to or on a portion of the shelving unit, wherein: the wireless-power receiver comprises an antenna and power-conversion circuitry that is configured to convert the RF wireless-power waves harvested by the antenna into usable energy for charging a battery of the small form-factor device or for powering the small form-factor device.
2. The wireless-power transmitting system of claim 1, wherein the wireless-power transmitter is coupled to a supporting structure of the enclosed environment and configured to radiate RF wireless-power waves towards a plurality of the one or more shelving units in the enclosed environment.
3. The wireless-power transmitting system of claim 1, wherein the small form-factor device is configured to at least one of transmit, receive, or display data.
4. The wireless-power transmitting system of claim 1, wherein the enclosed environment includes at least one of: a room, a building, a warehouse, a cargo compartment, or a delivery compartment.
5. The wireless-power transmitting system of claim 1, wherein the antenna of the wireless-power receiver is a stamped metal harvesting antenna coupled to a PCB that includes components for operating and powering the small form-factor device.
6. The wireless-power transmitting system of claim 5, wherein the stamped metal harvesting antenna is quasi omnidirectional and configured to receive RF wireless power waves having any

polarization.

7. The wireless-power transmitting system of claim 1, wherein one or more shelving units include one or more shelves and a respective wireless-power receiver is coupled to each one of the one or more shelves.

8. The wireless-power transmitting system of claim 1, wherein the wireless-power transmitter comprises a radio frequency (RF) wireless-power antenna that is circularly polarized such that the radiated RF power waves are transmitted having a circular polarization.

9. The wireless-power transmitting system of claim 1, wherein the first shape and the second shape are a meandering path that produces a rectangular shape.

10. A method, comprising: radiating, via a wireless-power transmitter coupled within an enclosed environment, RF wireless-power waves towards one or more shelving units within the enclosed environment, the wireless-power transmitter including: a ground plane, a first plurality of conductive segments having a first shape and a first size, a second plurality of conductive segments having a second shape and a second size, the second shape being distinct from the first shape and the second size being distinct from the first size, wherein a respective conductive segment of the second plurality of conductive segments separates adjacent conductive segments of the first plurality of conductive segments from one another, and the first and second pluralities of conductive segments being disposed over the ground plane to form an antenna that is configured to radiate RF wireless-power waves; wherein: the RF wireless-power waves are configured to be received by a wireless-power receiver coupled to or on a portion of a shelving unit of one or more shelving units, and the wireless-power receiver that is configured to convert the RF wireless-power waves, harvested by an antenna of the wireless-power receiver, into usable energy for charging a battery of a small form-factor device or for powering the small form-factor device.

11. The method of claim 10, wherein the wireless-power transmitter is coupled to a supporting structure of the enclosed environment and configured to radiate RF wireless-power waves towards a plurality of the one or more shelving units in the enclosed environment.

12. The method of claim 10, wherein the small form-factor device is configured to at least one of transmit, receive, or display data.

13. The method of claim 10, wherein the enclosed environment includes at least one of: a room, a building, a warehouse, a cargo compartment, or a delivery compartment.

14. The method of claim 10, wherein: the RF wireless-power waves are configured to be radiated using a first frequency band; and the wireless-power transmitter comprises a backplane including one or more data-communications antennas configured to transmit data signals using a second frequency band distinct from the first frequency band.
